



Comprehension Passage Pack for Grade 5

This resource contains the full text of reading comprehension passages in Levels 19 through 21 of Lexia® Core5® Reading. It supports teachers in further scaffolding comprehension instruction and activities for students.

The comprehension passages in Lexia Core5 Reading have been analyzed using a number of tools to determine complexity, including Lexile® measures. Based on this analysis, the comprehension passages are appropriately complex for students reading at the grade-level of skills in each program level. For example, the comprehension passages in Levels 19-21 (Grade 5 skills) typically fall within the range of Lexile measures deemed appropriate for on-level Grade 5 readers. (Texts with non-standard punctuation, such as poems and plays, are not measured.)

The *Content Area Connection* column in the table of contents can be used as a guide to determine the general topic of each passage. It does not indicate alignment to any specific content area standards.

Keywords in the passages are indicated in bold and defined in a glossary located at the end of the pack. The words are the same as those found in the online passages. While most terms are included to support word meaning, some terms are included because pronunciation may be challenging.

Reading Comprehension Passages: Levels 19-21

Passage Title	Genre	Content Area Connection	Lexile Measure	Page
Core5 Level 19				
Balancing the Needs of People and Plovers	Informational Text	Earth & Space Science	980L	4
An Uninvited Guest	Narrative Text	Earth & Space Science	870L	6
The Business of Zoos	Informational Text	Earth & Space Science	970L	8
When Lightning Strikes	Informational Text	Earth & Space Science	920L	10
"Hail" by J. Patrick Lewis	Poetry	Earth & Space Science	NA	12
Walls of Fire	Narrative Text	Earth & Space Science	870L	13
Join the North School Walkers	Opinion/Persuasive	Social Studies	880L	15
Support Art in the Park	Opinion/Persuasive	Social Studies	950L	17
We Can All Be Winners Here!	Opinion/Persuasive	Social Studies	880L	19
Borrowing Nature's Designs	Informational Text	Life Science	960L	21
Splendid Spiders	Informational Text	Life Science	900L	23
"The Secret Song" by Margaret Wise Brown	Poetry	Life Science	NA	25
A Review of Treasure Island	Opinion/Persuasive	English Language Arts	940L	26
A Movie for Everyone	Opinion/Persuasive	English Language Arts	810L	28
Just Listen to This!	Informational Text	English Language Arts	840L	30
Adventure Island	Narrative Text	English Language Arts	850L	32
The Harpies	Myth	English Language Arts	810L	34
an excerpt from <i>Five Children and It</i> by Edith Nesbit	Narrative Text	English Language Arts	790L	36
Core5 Level 20				
The Tunguska Blast	Informational Text	Earth & Space Science	930L	38
Mysteries of the Deep Sea	Informational Text	Life Science	910L	40
The Mystery of the Nazca Lines	Informational Text	Earth & Space Science	920L	42
The Maiden Wiser than the Tsar	Folktales	Social Studies	890L	44
Wealth and Worries	Drama	Social Studies	NA	46
an excerpt from "Songs for the People" by Francis Harper	Poetry	Social Studies	NA	49

Passage Title	Genre	Content Area Connection	Lexile Measure	Page
Core5 Level 20 continued				
The Travels of Marco Polo	Informational Text	Social Studies	920L	50
Dots and Dashes	Informational Text	Social Studies	960L	52
Propaganda or Truth?	Informational Text	Social Studies	830L	54
Dora, the Dog Wonder	Narrative Text	English Language Arts	920L	56
Rip Tide	Narrative Text	English Language Arts	910L	58
A Close Circle of Friends	Narrative Text	English Language Arts	880L	60
The Pollinators	Informational Text	Life Science	940L	62
Life Underfoot	Informational Text	Life Science	860L	64
Living with Good Germs	Informational Text	Life Science	930L	66
It Came From Space	Informational Text	Earth & Space Science	810L	68
Phaethon: A Greek Myth	Myth	Earth & Space Science	870L	70
The Moon: True or False?	Informational Text	Earth & Space Science	770L	72
Core5 Level 21				
Travels with Ray	Narrative Text	Life Science	920L	74
an excerpt from "Sol Painting, Inc." by Meg Medina	Narrative Text	Social-Emotional Learning	880L	76
an excerpt from "Secret Samantha" by Tim Federle	Narrative Text	Social-Emotional Learning	970L	78
Lonnie Johnson, Innovating from the Beginning	Informational Text	Engineering & Technology	920L	80
an excerpt from <i>Salt: A Story of Friendship in a Time of War</i> by Helen Frost	Poetry	Social Studies	NA	82
an excerpt from <i>Salt: A Story of Friendship in a Time of War</i> by Helen Frost	Poetry	Social Studies	NA	84
Glossary				86

LEVEL 19, UNIT 1
INFORMATIONAL TEXT

Balancing the Needs of People and Plovers

Certain kinds of **plovers**, such as the piping plover, hooded plover, and western snowy plover, build their nests on sandy beaches. They build them between dunes or sea walls and the high-tide mark. This is precisely where beachgoers like to lay down their towels to enjoy a day at the beach.

This has created quite a debate. On one side are the cute little birds that have been described as “cotton balls on toothpicks.” Their light brown, white, and gray coloring makes them hard to spot on the sand. In spring, these birds lay their tiny, sand-colored, hard-to-see eggs in shallow nests dug into the sand.

These nests face danger from many sources. Storms and **surging** waves may wash them away. The eggs may be crushed by careless humans (on foot, in off-road **vehicles**, and with dogs). They may also be eaten by **predators** (such as foxes, cats, gulls, crows, and ravens).

If the eggs survive and hatch, it takes over a month for the chicks to grow strong enough to fly. To help them grow, plovers look for food by the water’s edge or in seaweed on the beach. If they’re frightened by people or predators, they run and hide wherever they can. This running and hiding uses up valuable energy. If it happens often enough in a day, a chick will starve.

Because the number of beach-nesting plovers is so low, their status is “**threatened**.” This means there are laws to protect them and organizations looking out for them. Some of the **techniques** used to protect nesting plovers include putting ropes around nests and providing little wooden shelters for chicks to hide in. The most extreme of all is closing off part or all of a beach during nesting season (which occurs between April and August).



All this protection angers the other side of the debate: beachgoers. Plovers are drawn to exactly the wide, sandy beaches that people like to frequent. When it has come down to the plovers' right to protection or people's beach-going rights, plovers have won. People find themselves cut off from favorite beaches and crowded into what is left. But all this may be changing.

Plover experts now believe that the biggest dangers to plover nests are storms and predators. Nothing can be done about storms, and destroying or relocating plover predators creates new problems. But it does seem that predators stay away from beaches often filled with people. So some beach communities have begun to have a more "relaxed" attitude. They rope off any plover nests and post warnings to protect the plovers, but they don't close off the beach. The little birds seem willing to coexist with people, as long as the people don't disturb their nests or chicks.

Maybe, just maybe, we can all get along!

LEVEL 19, UNIT 2
NARRATIVE TEXT

The dark-winged, **unidentified** flying object swooped from the ceiling toward the family seated in the kitchen and back up again. Bashir screamed, and his sister Aisha slid off her chair to hide under the table. "Don't worry," their grandmother said calmly. "It's just a bat."

"What do you mean it's just a bat?" Bashir shrieked, waving his arms frantically to keep the winged creature away. "Those things carry rabies!"

Aisha peered out from under her hiding spot. "Babies? I don't see its babies."

Bashir shook his head and answered, "Not babies—rabies, rabies! It's a disease you get from bats. They bite you, and then you die."

Aisha started to cry, and their grandmother said, "Let's not overreact. First, very few bats carry rabies. Second, there's a medical treatment for rabies. And third, this poor bat is probably just as afraid of you as you're afraid of it."

"Who's afraid?" asked Bashir in the bravest voice he could **muster**. Just then, the bat spread its wings and glided toward Bashir's head. He screamed again and ducked behind Aisha under the table.

"I heard that bats like to get tangled in people's hair!" Aisha **shuddered**.

"That's only a myth," **reassured** their grandmother as she quickly gathered a blanket that had been draped over a chair. "And this poor animal will soon exhaust itself." Sure enough, after a few more swoops around the room, the bat finally settled on a high shelf. A quick toss of the blanket succeeded in trapping the bat in its soft folds.



"You caught it!" exclaimed Bashir. Every time he visited his grandmother, he was always impressed that she knew exactly how to handle any situation.

Bashir crept closer to the bundle in his grandmother's arms and saw that the bat's eyes were fixed on him with a combination of what he thought might be curiosity and fear. "Now what should we do with it?" he asked in a hushed tone.

"We need to set it free, of course," their grandmother asserted, walking to the door and opening it. "I'm not sure if this bat wants to make a home in our attic or if it's lost, but it will be much more comfortable outside." She shook the blanket gently to release the bat, and they watched it soar into the distance until it disappeared.

Bashir breathed a sigh of relief, and his grandmother smiled.

"I'm always happy to see bats flying around *outside* because they **devour** mosquitoes by the thousands," she remarked. "Without bats, the world would be much buggier than it is. We need them, even though we don't like to share our indoor spaces with them."

One evening not long after the weekend visit with his grandmother, Bashir noticed familiar black figures circling **swiftly** against the dimming sky outside his apartment window. He recognized them immediately and murmured, "Hello, my mosquito-eating friends," pleased that this time he did not feel afraid at all.

LEVEL 19, UNIT 3
INFORMATIONAL TEXT

The Business of **ZOOS**

Early in the twenty-first century, **conservation** groups in **Thailand** **protested** against shipping elephants to zoos in Australia. Supporters of animal rights in the United States claimed that elephants in city zoos were being harmed. The leader of one animal-protection group asked, “Is there any value having elephants at zoos other than to allow people to see them in person?”

Allowing people to see **exotic** animals is a main purpose of zoos. Ever since ancient times, wild animals from distant lands have been put on display, simply for people’s viewing pleasure.

A zoo animal was placed in a cage with bars. Keepers fed it and cleaned its cage, but paid no attention to its other needs. A caged animal had nothing to do. Zoo visitors might see a lion or a bear endlessly **pacing** in its tiny cell. They might see a gorilla sitting on a concrete floor, staring blankly. At times, visitors felt more sadness than awe.

It was not until recently that many zoos began to change. Zoos created natural-looking environments, such as rainforests and large outdoor enclosures. Many zoos offered animals more space and stimulating activities. But zoo critics point out that, even in natural-looking environments, most animals have nowhere to hide, as they would in nature. They must be on display for visitors.

Some opponents of zoos say that there is no need for people to see exotic animals up close anymore. Television and the Internet make it easy to view wild animals in their natural **habitats**.



Supporters of zoos argue that educating the public about wildlife is a worthwhile goal. When people get close to a mighty gorilla or an adorable panda, for example, they are willing to give money to help save these animals' threatened habitats. The people who work at zoos are often active in conservation groups that protect species in the wild. Many zoo doctors have saved the lives of sick animals in wildlife preserves.

Zoo supporters also highlight that today's zoo animals are mainly born in **captivity**. The animals are not caught in the wild and brought to the zoo. In captive-breeding programs, baby animals are born in zoos and then returned to the wild one day. For example, the breeding program set up by a New York zoo in the early 1900s saved an animal called the American **bison** from **extinction**. Today, tens of thousands of American bison roam the North American plains. Captive-breeding programs tend to be costly, however. And while they were successful at saving American bison from dying off, they have not worked as well to save other animals. To survive in the wild, most animals need to be born in the wild.

Very few people want to ban zoos entirely. Zoo critics and supporters alike, including many in the zoo business, have some shared goals. They want to make sure that captive animals live well, and that wild animals can live freely.

LEVEL 19, UNIT 4
INFORMATIONAL TEXT

When Lightning STRIKES

Imagine two huge lightning bolts **simultaneously** strike the tips of two skyscrapers. Impossible? Well, just such a thing happened in Chicago in 2010, and one photographer was lucky enough to capture the spectacular moment. But then, lightning flashes often have a way of being highly dramatic.

Most of us don't see many lightning flashes in a year, but don't be fooled. According to recent satellite data, over three million lightning flashes occur worldwide every day. Most travel from cloud to cloud, but about 860,000 of them strike either the ground or some water surface on Earth.

Lightning travels at the speed of light, which is 186,282 miles per second (299,792,458 meters per second). The reason the thunder we hear trails far behind the lightning we see is that the speed of sound is **comparatively** slow. It takes five seconds just to travel a mile in warm summer air. This difference in speeds provides a quick way of estimating how close an electrical storm is. As soon as you see a lightning flash, start counting seconds (*one thousand and one, one thousand and two*). Stop when you hear the thunder and divide by five to get the number of miles.

As enjoyable as lightning is to watch, it can do tremendous damage. Lightning strikes are the major cause of forest fires and frequently cause power **outages**. A lightning strike in northern New York caused a blackout that **paralyzed** New York City in 1977. More importantly, about 24,000 people are killed by lightning every year. Ten times that number are seriously injured. So if you see a flash of lightning, start counting. The latest guidelines say to head for shelter as soon as that number is under 30.



If you're in a car, make sure the windows and doors are closed. If you're outdoors and can't reach a building, avoid anything tall in your area. Lightning tends to take the most direct route to Earth, striking the closest (tallest) object that happens to be in its path. Stay away from single trees, high fences, and other such structures, especially metal ones that **conduct** electricity. Avoid open areas. If you can't get out of the open, crouch close to the ground. If you are swimming in water, get out.

If you make it indoors, you still have to be careful. These days, buildings include various forms of lightning protection, but lightning is tricky. It can travel through phone lines, so only use cell phones for calls. It can come through faucets, so don't take a shower or wash anything during a storm. Don't stand close to windows.

Electrical storms are amazingly beautiful, but don't forget that they are also amazingly dangerous!

LEVEL 19, UNIT 5
POETRY

The hail flies
on furious hooves.

It batters cars
and rooftops,
slamming anger,
and then melts away.

– J. Patrick Lewis

LEVEL 19, UNIT 6
NARRATIVE TEXT

WALLS OF FIRE

"Alana, is everything going to be okay?" Lidia asked me. We were holding hands tightly in the **rec center**, waiting for rescuers. Not long before, we'd been swimming in the pool with other campers when the camp director suddenly appeared. She announced that everyone had to get out, get dressed, and follow the counselors to the rec center—fast!

Inside the wooden building, we sat on the floor while the director told us, "We must **evacuate** the camp because there's a fire in the woods. Buses are coming to drive us out of here, but we have to wait for them to arrive."

That's when I realized that the burning **sensation** in my eyes didn't come from the pool water, but from smoky air. Gray-black clouds swirled above distant treetops.

The rec center had a TV, and someone turned it on. We watched a live news report about a **wildfire**. The cause was unknown—maybe a lightning bolt or a stray spark from a campfire. The long **drought** in the region meant that dry leaves, grass, and sticks caught fire easily. Then the wind spread the flames, and walls of fire rose high into treetops. Video showed teams of firefighters with chainsaws, shovels, and other tools, laboring in the hot-oven forest to clear **vegetation** and control the blaze. Yellow-orange flames crept and leaped in the background.

The reporter pointed to three parked buses, explaining that they could not get through to a children's camp because of burning trees that had fallen across the dirt road. "The children are trapped," he said.



The director turned off the TV, but we had already seen too much. Those trapped children were US! Trapped! Some of the campers were crying. Fear grabbed me by the throat, and I gasped for breath.

The nature counselor tried to calm everyone down by telling us about wildfires—how common they were and how some trees even depended on fire to spread seeds. I wondered if I'd ever see my family again.

Waiting was **agonizing**. The air thickened with smoke. Harsh odors filled our nostrils.

Two long hours later, flashes of color appeared through the haze. Had the flames reached the camp? No, three yellow school buses were on the road. We cheered as our rescuers pulled up in front of the building.

On the drive out of the camp, we could barely see the woods through the smoke. And when we finally reached the open highway, it was like **emerging** from a dark, deadly, **alien** world.

This happened six years ago, but the events are burned in my memory. It was the scariest day of my life!



LEVEL 19, UNIT 7
OPINION/PERSUASIVE TEXT

JOIN THE NORTH SCHOOL WALKERS

Dear Parents of North School Students:

Have you heard about our walking-school-bus program? It began last year with 30 students. As the new school year begins, 50 students have enrolled. It would be great to see 100 or more students participate, because this program has many benefits that will help our children and our community.

Here's how the program works. Your child waits for the "bus" each morning at an **assigned** time and place. The bus is not a **vehicle**, but a group of students with one adult leading and one adult following. All the walkers stay together, with the adults making sure that everyone crosses streets safely. At the end of the school day, the walkers reverse their route. Right now, the adults are volunteers, but we hope to find the money to pay workers soon.

Even with paid workers, the program saves money for our school **district**. Because of sharply rising fuel costs, school buses take a big bite out of our district's **budget**. Fewer buses mean lower costs. Saving money to use for education instead of **transportation** is a major reason for this program, but there are other reasons, too.

The walking-school-bus program is good for everyone's health. Health experts point out that today's youngsters are not getting enough exercise, and are at risk of developing health problems as a result. Walking is a simple and effective physical activity. And it doesn't pollute the air with vehicle fumes!



In addition, students have a safer way to get to school. The walking-school-bus program **reduces** not just the number of school buses, but also the number of parent-driven cars that **clog** the streets by North School. When students no longer have to dodge traffic to enter the building, safety will be improved.

There has been an unexpected **bonus** to the program—a greater sense of community among the walkers. What better way to feel part of a neighborhood than by taking a daily walking tour! Help your child and our community by enrolling in the North School Walkers program. Sign-up sheets are in the school office.

Sincerely yours,

Edda Freeman and Tony Palermo

Parent Coordinators of the North School Walkers



LEVEL 19, UNIT 8
OPINION/PERSUASIVE TEXT

Support

Art in the Park

Dear Editor,

I'm writing to urge everyone to come to Art in the Park this Saturday in Broad Creek Park, from 9 AM to 6 PM. There is no admission fee, and there's so much to do and see. Over 150 local artists will be showing their work, hoping to sell some of it to you. There will also be an art show featuring the best work of fifty students from local schools. The student works have already been judged, so you can see if you agree with the judges' choices. They have awarded blue, red, and white ribbons to some of the up-and-coming artists in our local high school, middle school, and two elementary schools.

This year, we are introducing a special section set aside for even younger artists. It is called Kids in the Park. It comes equipped with fingerpaints, chalk, watercolors, crayons, and lots of paper so kids can make their own artistic creations. And don't worry about mess. We provide smocks for kids to wear to keep their clothes clean!

A local band, the Snow Squalls, will be providing country music to keep things lively. And a food court, run by **Aroma** Market, will have a wide variety of scrumptious treats. (As everyone knows, Aroma Market is famous for its tasty food!)

This is the third year for Art in the Park. We have been encouraged by the number of people who attended the first two times, but we need more of you to come! Supporting art in our community has so many benefits for you, for your kids, and for our local artists.



It's a chance for you to get to know other community members as you stroll through the show. It's a chance for you and your kids to experience different kinds of art: paintings, photographs, sculpture, pottery, **carvings**, jewelry, and more. It's a chance for you to discuss what you see with the artists themselves. Don't be afraid to ask questions or to share opinions. Talking with community members is one of the main reasons the artists enjoy participating in this event!

So imagine yourself, strolling through our beautiful park, listening to entertaining music and eating delicious food. As you view a wide range of art, one piece in particular catches your eye. You have an interesting chat with the artist and then you decide to buy the piece. It didn't cost that much, and you know it will look just beautiful in your home.

Now stop imagining, and experience the real thing! Support and get to know your local artists, encourage student artists, help your community, and enjoy yourself.

We hope to see you this Saturday!

Antony Alvarez

Art in the Park Chairman

LEVEL 19, UNIT 9
OPINION/PERSUASIVE TEXT

We Can All Be Winners Here!

By Lamar Green, Grade 6

If you listen to the news these days, you know that everybody is concerned that kids aren't eating enough nutritious food. If any of these concerned **citizens** ever visited our cafeteria at lunch time, they might worry even more! Yes, the daily hot lunch features healthy food, but the vegetables taste terrible. They're either overcooked frozen vegetables, or if they're in salads, they are not very fresh. And then what happens? Kids leave the vegetables untouched. Who wants to eat mushy broccoli or **wilted** lettuce? So the school spends money on food that gets thrown away, and we kids don't get a complete, nutritious meal. Everybody loses.

But there's a way for everyone to win: Our school could set up a farm-to-school program. Over 2,000 schools all across the country already have farm-to-school programs that are running successfully. Here's how it works: The school sets up an arrangement with local farmers, who sell certain fresh fruits and vegetables to the school every few days. Farm-to-school programs also include classroom visits from farmers, who help kids understand what's involved in farming, and field trips for kids to see real farms firsthand.

The farmers win because they have a nearby market for their crops that they can depend on. They don't have to drive around, selling their food to different stores at a lower price, so the stores can turn around and sell it at a higher price. This means farmers earn more money and don't have to spend as much on transportation.



The school wins, too, because it gets to know the local farmers. It can work with farmers to decide which kinds of fruits and vegetables to grow. (Maybe we kids can make suggestions, too—that would be cool!)

But most of all, kids win. We get tasty, fresh fruits and vegetables to eat. Think how fantastic it would be if our cafeteria also included a great salad bar! And along with eating better, we could learn more about where our food comes from through classroom visits and field trips. We'd get to know the farmers in our community and see for ourselves how our food is grown.

I hope I've made it clear why our school should set up a farm-to-school program. Everybody wins!

LEVEL 19, UNIT 10
INFORMATIONAL TEXT

Borrowing Nature's DESIGNS

Imagine a day when doctors could help a person grow a replacement body part—just like starfish and lizards can do. What if bridges were built from something much stronger than steel—fibers made of spider silk! Some **engineers** study nature's designs in order to develop new technologies. These engineers are using a kind of science called biomimetics. *Biomimetics* comes from the Greek words *bios* "life" and *mimesis* "to imitate."

Prickly Plant Parts

In 1941, a Swiss engineer named George de Mestral went for a hike with his dog. Afterward, picking off burrs from his clothes and dog's fur, he became curious. How did the **burrs** manage to stick so well? He studied the **structure** of each burr and discovered that the spines ended in tiny hooks. He went on to invent a kind of fastener made of tiny hooks. He helped start a company to **produce** the fasteners. Velcro® is still a trade name for these burr-inspired fasteners. They are used on clothing, shoes, school binders, and many other products.

Sticky Feet

Little lizards called geckos have an astounding ability to cling to walls, ceilings, and other **surfaces**. For decades, scientists have studied gecko feet to figure out what makes them so sticky. Experiments show that millions of microscopic hairs on each toe are mainly responsible for a gecko's grip. **Molecules** on the hairs interact with molecules on the surface. The resulting **forces** pull the toes and surface together. But geckos don't just stick to a surface—they unstick, too. To do that, a gecko tugs the foot in the opposite direction, releasing the grip.



Like Velcro,[®] products that can stick firmly yet be removed easily have practical uses. So far, engineers have developed gecko-inspired robots that climb walls. They've made a fabric-like material that can stick to smooth surfaces. They're also working on grippers that might someday be used to pick up junk floating in space.

Rough Skin

Sharks are extremely speedy swimmers. Any solid body moving through water, including a shark, encounters a force called drag. Drag slows moving objects down. A shark's skin reduces drag, helping it be such a fast swimmer. Powerful microscopes reveal that sharkskin is covered in tiny toothlike structures called denticles. Denticles cut through the water, helping sharks move more smoothly through the water and swim faster. Engineers designed materials with denticle structures to make swimsuits first used by Olympic swimmers in 2008. Denticles also prevent **organisms** like **algae** from attaching themselves to the shark's skin. Engineers are experimenting with denticle structures to create a material that **repels** bacteria. If it works, hospitals may coat surfaces with this material to reduce infections.

What problems will biomimetic engineers try to solve next? One thing is for sure: nature will provide inspiration for new solutions.

LEVEL 19, UNIT 11
INFORMATIONAL TEXT

Splendid Spiders

Fear of spiders is common. “Ugh!” people say, “Keep that creepy thing away from me.” Many people get out the broom when they see a spider in the house. And some people panic when they see just a photo of a spider. Fearful **reactions** may be common, but they are not sensible. Only a few kinds of spiders can harm people. Spiders are actually helpful because they eat insects. Spiders are fascinating creatures that deserve to be admired.

Spiders belong to a class of animals called **arachnids**. Unlike insects, spiders have eight legs, not six, and no wings or **antennae**. There are more than 35 thousand known **species** of spiders, with more species yet to be discovered. Spiders are successful **predators** that live all over the world.

Spiders are one of the only organisms with the ability to produce silk from their bodies. About half of the world’s spiders use their silk to spin webs, which are highly effective traps. The spider rests quietly in the web or nearby, waiting for its **prey**. An insect that lands on the sticky strands of silk cannot escape. Webs come in many shapes, including funnels, sheets, and messy-looking cobwebs. The most familiar image of a web belongs to **orb**-weavers. Orb-weavers, such as the garden spider, create large, delicate, and beautifully patterned webs that **glisten** with dew.

Spider silk is famous for its strength and its ability to stretch without breaking. The very large webs of certain orb-weavers have even been used as fishing nets. A spider produces different kinds of silk for different purposes. Dragline silk, for example, is a lifeline for a dangling spider. It is stronger than a steel wire of the same width, and much more stretchable. **Engineers** and scientists study spider silk as they try to make a fiber that is equally strong and flexible.



Such lab-made spider silk could have many uses, from ultra-strong fabrics to supports for broken bones.

Spiders do amazing things. For example, they taste their food by using the hairs on their legs. They digest their food before they swallow it, using chemicals to turn it into liquid. Young spiders leave their birthplace by “ballooning”—riding air **currents** on lightweight silk threads. In addition to spiders that trap prey, there are spiders that jump, spiders that spit, and spiders that fish. Trapdoor spiders live in silken **burrows** with removable lids. When an insect passes by, the spider pops out from under the lid and grabs its prey in a flash.

People may say, “A spider—ugh!” However, once they learn a little more about spiders, it might be more fitting to say, “A spider—wow!” Spiders are marvels of the natural world.

LEVEL 19, UNIT 12
POETRY

The Secret Song

Who saw the petals
drop from the rose?
I, said the spider,
But nobody knows.

Who saw the sunset
flash on a bird?
I, said the fish,
But nobody heard.

Who saw the fog
come over the sea?
I, said the pigeon,
Only me.

Who saw the first
green light of the sun?
I, said the night owl,
The only one.

Who saw the moss
creep over the stone?
I, said the gray fox,
All alone.

- Margaret Wise Brown

LEVEL 19, UNIT 13
OPINION/PERSUASIVE TEXT

Treasure Island

by Robert Louis Stevenson

Reviewed by Ellison Shih, Grade 8

The book *Treasure Island* is on every list of classic adventure stories. This book is described with words like *suspenseful*, **rousing**, and *fast-paced*. Robert Louis Stevenson wrote the book a long time ago. It was published in 1883. I wondered whether it would still seem as exciting as people say. I'm a big fan of adventure stories, so I decided to read the book for myself.

The story takes place in the 1750s. The narrator is 12-year-old Jim Hawkins, who has found a map to a treasure that pirates buried on a distant island. Jim serves as a cabin boy on a ship sailing to the island. He is **befriended** by the ship's cook, Long John Silver, who is none other than the cruel leader of the pirate gang that is pretending to be the regular crew on the ship. The author drops clues to what is really going on. But Jim realizes the truth only after he **overhears** the pirates making their plans to find the treasure and take over the ship. He escapes to the safety of a wooden fort on the island, along with the captain, the ship's owner, the doctor, and a few honest sailors.

In the following battles against the **mutineers**, Jim is an action hero, brave, lucky, and clever. He performs daring **deeds**, and more than once **outwits** his **scheming**, heartless enemy.

The character of Long John Silver is one of the most interesting in the book. He is a mean-spirited pirate, who is still able to charm others. He can talk his way out of trouble and find opportunity in situations that seem hopeless.



Sometimes the language of the book is hard to understand. The characters use words and expressions that are not spoken anymore, and there are many sailing terms that I didn't know. But I understood enough to follow the action, and action is the most important feature of the book. As I read, I kept thinking that the story would make a great movie or graphic novel.

It turns out that *Treasure Island* has been made into a movie more than once. I plan to watch a movie version, because it's always interesting to compare a book and a movie that tell the same story.

On a scale of 1 to 10, I give the book *Treasure Island* a 7. It is old-fashioned in some ways, but I did want to keep reading to find out what would happen next.



LEVEL 19, UNIT 14
OPINION/PERSUASIVE TEXT

A Movie for **Everyone**

by Riya Sharma

Are you looking for a film to rent that the whole family can enjoy? I've got the perfect movie for you. It has something for everyone: adventure, suspense, comedy, tragedy, and best of all, singing and dancing. Even if you don't think you like movie musicals, give it a try. I don't like most **musicals**, but I love this movie.

The name of the movie is *Oliver!* (The exclamation mark isn't from me. It goes with the title.) It was released in 1968 and went on to win a raft of awards, more than most films ever will. Since then, audiences of all ages have been happily viewing it again and again. The movie is based on *Oliver Twist*, a classic book by the British writer Charles Dickens.

Oliver! sticks to the part of the story that tells about *Oliver Twist's* childhood, which takes place in the early 1800s in England. The boy, who is an **orphan**, gets swept up in a gang of kids who are thieves and **pickpockets**. These kids are tough, but they aren't mean. They really look out for each other. Their leader is an outrageous old man named Fagin. He's not so nice, but you'll love to hate him! And he'll have you helpless with laughter as he instructs the boys, singing "You've Got to Pick a Pocket or Two."

Almost as memorable is a gang member called the Artful Dodger. (His nickname means that he's quite good at getting himself out of danger.) Dodger welcomes Oliver into the group with a song, telling the orphan to "consider yourself part of the family." He then becomes Oliver's loyal friend and protector.



I mention some of the songs because they, and the other great songs, fit so well with the story. You'll want to sing along with these powerful tunes, so I strongly recommend going online to find and print out the song lyrics. After all, you'll be sitting in your own home, not bothering anyone else. Why not sing along? And while you're at it, enjoy the fantastic dance sequences as well.

Let me add just a few words of caution. If really young children are watching, tell them that everything turns out all right for young Oliver. Otherwise, they might get upset by some of the things that happen, especially when nasty Bill Sykes is around.

But *Oliver!* isn't meant to be a frightening movie. It's really a celebration of love and friendship, full of fun and energy. It has interesting characters that little kids will like and that older kids and parents will get a kick out of. After you see *Oliver!* you'll know why they added that exclamation mark to the title!

LEVEL 19, UNIT 15
INFORMATIONAL TEXT

Just Listen to This!

How often do you listen to classical music? If you're like most young people, your answer is "Never." Well, you're missing out on a terrific listening experience. Give classical music a try and you may find that you like it.

The term *classical* describes a style of music that was developed in Europe in the 1700s and 1800s. One kind of longer classical work, called a **symphony**, is composed for the instruments of an **orchestra**. If you want to listen to a symphony for the first time, find a recording of Symphony Number 6 by one of the greatest composers (some say the greatest) who ever lived: **Ludwig van Beethoven**.

The reason to start with the Sixth Symphony, also known as the Pastoral, is that Beethoven suggested feelings and images for each movement. (A movement is a separate work within the symphony.) Each of the five movements presents a scene that listeners can picture, so as you listen, try to imagine the setting. How does the music convey the mood and the action? Listen for repeated **melodies**, called themes. Listen for how the different sections of the orchestra play with and against each other. Listen to how the music grows softer or louder to express gentle or strong feelings. Try to identify the different solo instruments you hear.

The first movement of the Sixth Symphony brings listeners to the countryside. Beethoven loved the country, and the music conveys calm and happy feelings. Listen for the solo **clarinet** and you will feel the pleasure of being in this beautiful place.

The second movement takes place by a brook and you may hear the **murmuring** water of the **stringed instruments**. Listen carefully to see if you can hear the calls of birds. There is a flowing, dreamy feeling to this movement.



In the third movement, the country folk gather for a lively dance. Picture the joyous whirling of the dancers as you listen. Next, something arrives to disturb the dance: a thunderstorm!

In the fourth movement, drums make rumbling thunder. Listen to the **trombones** sound like rain and imagine the darkening skies with everyone hurrying for shelter. You can even hear the high notes of lightning.

The melody of a shepherd's song ends the fourth movement and begins the fifth. The storm is over. If you've ever wondered what sunshine sounds like, just listen to this music. The glorious sun lights the world again, and everyone gives thanks.

Beethoven completed his Sixth Symphony in 1808. He had begun to lose his hearing six years earlier, suffered from loud ringing in his ears, and his deafness was increasing. Yet, this genius was able to hear this wonderful music in his "mind's ear," and share it with listeners for all time.

Listen, and enjoy!

LEVEL 19, UNIT 16
NARRATIVE TEXT

David's class was studying maps. For homework, each student was supposed to find a map of an interesting place to share. David wished he could show Adventure Island. He wouldn't be able to find a map of it, though, because it wasn't a real place but one he sometimes daydreamed about.

David was getting ready for bed when a great idea came to him. Instead of finding a map, he could make one! He placed a large sheet of paper on his bedroom floor and began to draw a map of Adventure Island.

Adventure Island was, of course, surrounded by water, so David showed it in the middle of Lake Wonder. He drew Golden Beach, which had glittering sand, and Roller Coaster Hill, which was fun for bike riding, and Pirate Cove, where treasure lay buried. There was a blank area in the northeast, and David closed his eyes, trying to picture what to put there.

Suddenly, David heard the sound of flowing water. It grew louder. With a startling WHOOSH, David was no longer in his bedroom, and night had turned to day. He looked down to see his bare feet standing on mossy rocks. He looked up to see a waterfall spilling over a rock wall to a pool of water below. Without **hesitating**, he slipped into the cool water and began swimming toward the waterfall with strong, sure strokes. David had only learned to swim that summer and was still at the **doggy-paddle** stage in the town pool. But this water was different—it seemed to **propel** him, and he was hardly making any effort at all.



He felt the waterfall beating on his back, pushing him underwater, but he wasn't frightened. He wriggled like an eel toward a rocky ledge behind it. He climbed up on the ledge and peered through the curtain of water at the wondrous sunlit pool and the rolling green hills beyond. He almost couldn't believe that he had arrived on Adventure Island, but here he was, and it was even better than he had imagined!

A fish with silvery scales darted back and forth under the ledge, and David bent to get a closer look. It moved its lips in a fishy way and spoke out loud. "You must..." it said, but David couldn't make out its words. "You must..." it repeated, then louder, "You really must..."

The waterfall **vanished**, and David was lying on his bedroom floor. "You really must get into bed now," said his dad. "It's past midnight. The floor isn't a comfortable spot for sleeping."

As David got into bed, Dad asked, "Why is your hair so wet?"

LEVEL 19, UNIT 17
MYTH

THE HARPIES

A GREEK MYTH RETOLD

The brave hero Jason set out on a **quest** to find the Golden Fleece, the wool of gold from a magical **ram**. He journeyed with a group of other heroes, all seeking adventure. Among them were the sons of the North Wind: two mighty warriors with the gift of flight. Jason, the winged sons of the North Wind, and the other heroes sailed on their magical ship, Argo. They were known as the **Argonauts**.

On their quest, the Argonauts voyaged to many strange lands and **encountered** extraordinary creatures. This is the tale of their adventure in the land of King Phineus.

It was **twilight** when the Argo arrived on the shores near Phineus' kingdom. When the Argonauts reached the palace doors, they were met by an old and weak man who could not see. The blind man walked on trembling legs to greet them. He seemed barely alive, as if made of skin pulled over **jutting** bones.

The Argonauts were surprised to learn that this man was, in fact, King Phineus himself. "I am starving, though I have plenty of food," Phineus told them. "I have waited for your help, knowing that you would come." Jason and the Argonauts were eager to hear more. Why was King Phineus weak and starving? How did he know they would come?

King Phineus told his story. The god Apollo had granted Phineus the gift of seeing into the future. With Apollo's gift, Phineus knew what was to come. He used his new skill to warn people of future danger. Zeus, the king of all gods, was not pleased.

Phineus explained, "Zeus grew angry with me for telling **mortals** what was to come. He punished me, and his punishment has turned me into the starving, miserable person you see. Watch as I try to eat."



The king's servants laid out a feast. But as soon as Phineus reached for the food, terrible winged monsters with long claws and knife-like teeth dived from the sky and devoured almost everything. Even worse than their **savagery** was what they left behind—a smell so repulsive it was unbearable. The few remaining food scraps were utterly spoiled and reeking with the scent.

"What are those monsters?" asked Jason. Phineus' eyes grew wide. "Those hideous creatures—they are Harpies."

Phineus turned to the Argonauts and said, "Recall Apollo's gift to me: the gift of viewing the future. I have **foreseen** that among you are the ones who can save me from the Harpies. Please help and perhaps I can help you in return."

The Argonauts took pity on Phineus. But which hero among them was **destined** to defeat these winged monsters? Not one hero, but two, stepped forward eagerly. The two sons of the North Wind were determined to save the king. They drew their swords and pledged to drive the Harpies away. The wings on their ankles fluttered, ready to take flight.

Food was once again placed before Phineus. As before, when the old king brought a bite to his lips, the Harpies swooped down.

This time, however, the sons of the North Wind flew up after the Harpies, striking at them with their swords. The Harpies fled their fierce attackers, barely escaping with their lives. This moment was the last time the Harpies **tormented** the old king.

That night, Phineus and the Argonauts feasted undisturbed. In return, Phineus gratefully used his ability to see the future to tell the sailors how to survive the next dangers on their voyage. With this knowledge, the Argonauts awoke the next morning to continue on their search for the Golden Fleece.

LEVEL 19, UNIT 18
NARRATIVE TEXT

Five Children and It: An Excerpt

The classic novel *Five Children and It* tells about brothers and sisters who are visiting a country house in England. In this **excerpt**, Cyril, Anthea, Robert, and Jane are digging in a nearby gravel pit just for fun, when Anthea excitedly reports that she has heard a voice underground say “Let me alone.”

Then Anthea cried out, “I’m not afraid. Let me dig,” and fell on her knees and began to scratch like a dog does when he has suddenly remembered where it was that he buried his bone.

“Oh, I felt fur,” she cried, half laughing and half crying. “I did indeed! I did!” When suddenly a dry husky voice in the sand made them all jump back, and their hearts jumped nearly as fast as they did.

“Let me alone,” it said. And now everyone heard the voice and looked at the others to see if they had too.

“But we want to see you,” said Robert bravely.

“I wish you’d come out,” said Anthea, also taking courage.

“Oh, well—if that’s your wish,” the voice said, and the sand stirred and spun and scattered, and something brown and furry and fat came rolling out into the hole, and the sand fell off it, and it sat there yawning and rubbing the ends of its eyes with its hands.

“I believe I must have dropped asleep,” it said, stretching itself.



The children stood round the hole in a ring, looking at the creature they had found. It was worth looking at. Its eyes were on long horns like a snail's eyes, and it could move them in and out like telescopes; it had ears like a bat's ears, and its tubby body was shaped like a spider's and covered with thick soft fur; its legs and arms were furry too, and it had hands and feet like a monkey's.

"What on earth is it?" Jane said. "Shall we take it home?"

The thing turned its long eyes to look at her, and said—

"Does she always talk nonsense, or is it only the **rubbish** on her head that makes her silly?"

It looked **scornfully** at Jane's hat as it spoke.

"She doesn't mean to be silly," Anthea said gently; "none of us do, whatever you may think! Don't be frightened; we don't want to hurt you, you know."

"Hurt me!" it said. "Me frightened? Upon my word! Why, you talk as if I were nobody in particular." All its fur stood out like a cat's when it is going to fight.

"Well," said Anthea, still kindly, "perhaps if we knew who you are in particular we could think of something to say that wouldn't make you **cross**. Everything we've said so far seems to have. Who are you? And don't get angry! Because really we don't know."

"You don't know?" it said. "Well, I knew the world had changed—but—well, really—do you mean to tell me seriously you don't know a **Psammead** when you see one?"

LEVEL 20, UNIT 1
INFORMATIONAL TEXT

The Tunguska **Blast**

On the morning of June 30, 1908, a violent blast shook a forested region of central Siberia. From a distance, **witnesses** saw the sky in flames and heard loud booms. One man later reported being tossed out of his chair and feeling as if his shirt were on fire because the heat was so intense. A woman described how a powerful wind carried off the reindeer-**hide** cover of her family's tent. Her sleeping bag went up in flames. Afterwards, she saw that the forest was gone.

Instruments around the world detected shock waves from the blast. Dust from the explosion lifted high into the **atmosphere**, scattering sunlight that could be seen in far-off Europe at midnight.

War and other difficulties prevented scientists from reaching the blast zone. The first researchers did not arrive until 1927. They saw that 800 square miles (2,150 square kilometers) of forest had been leveled. Tens of millions of trees lay on the ground, all facing away from a central point. There were no signs of an earthquake or a volcano, so a likely cause was an object from space crashing into Earth. But why was there no **crater** marking the place of impact?

For more than 100 years, theories have been offered to explain the **Tunguska** event, named for a river of the region. Native people believed that a god who sent fire and thunder had struck the forest. Science fiction writers imagined that a spacecraft from another planet had exploded. Scientists proposed various ideas, including underground gases that **erupted**.

In 2007, researchers claimed that they had discovered a crater made by a space rock. The crater was now a lake. It was about 5 miles (8 kilometers) from the **epicenter** of the blast. Other scientists disagreed, saying the area around the lake did not have the features of an impact crater.



Today, the generally accepted explanation is that a chunk of an asteroid, or possibly a comet, entered Earth's atmosphere at an extreme speed, creating extreme heat. It blew up over the Tunguska region, releasing many times more energy than a nuclear bomb. Because the blast occurred above Earth's surface, there was no impact crater.

Asteroids are rocky bodies that orbit the Sun. Some come near Earth. Space agencies **monitor** Near-Earth asteroids that are at least 0.6 miles wide (1 kilometer), because an impact could cause global damage. So far, none of these large asteroids poses a threat.

The Tunguska blast, the largest in recent history, may have been caused by a fairly small asteroid, only about 164 feet across (50 meters). On average, asteroids of this size enter Earth's atmosphere every few hundred years. Another Tunguska-like event is sure to occur, though two obvious questions remain: *When?* and *Where?*

LEVEL 20, UNIT 2
INFORMATIONAL TEXT

Mysteries of the Deep Sea



Tales have long been told about kingdoms under the sea, which could be visited only by magic. The deep ocean was a mysterious **realm**, closed to real humans.

Fishing nets brought up strange-looking creatures from the depths, but nobody knew how these animals lived in their natural habitat. In 1934, the world's first submersible, or deep-sea diving **vessel**, dropped to the remarkable depth of 3,028 feet (924 meters). The two explorers in the submersible looked out the window and saw astonishing things.

Scientists are still learning about the astonishing life forms of the deep sea. Today's research submersibles can go deeper than ever before. At great depths, there is not even a glimmer of sunlight. The water is icy cold, and its **pressure** is body-crushing.

In the dark ocean depths, lights flash and glow. The lights come from the bodies of jellyfish, squid, fish, and other living things. The lights serve different purposes. Some animals may suddenly light up to **startle** a predator. Others may flash to **attract** prey, or a **mate**.

Green plants are the base of Earth's food chains, but without sunlight, no green plants can grow in the deep ocean. Animals here depend on "marine snow," tiny bits of dead plants and animals that drift down from the upper ocean layers. As with all food chains, the eaters become prey for other eaters.

The viper fish is a fearsome-looking predator. Its oversized jaws are filled with sharp teeth that curve inward. A full-size viper fish is only about 12 inches long (30 centimeters). But other deep-sea creatures are giants.



Oarfish live about a mile (1.6 kilometers) below the ocean surface. They are the longest bony fish in the world, up to 36 feet long (11 meters) and weighing 600 pounds (272 kilograms). It's possible that early sightings of giant oarfish led to stories about sea serpents.

Unlike the oarfish, the giant squid is an invertebrate, an animal with no backbone. Like the oarfish, the giant squid is probably the source of tales of sea monsters. The giant squid is nearly 60 feet long (18 meters)! Each of its enormous eyes is the size of a human head. In spite of its huge size, this deep-sea dweller is hard to find. A living giant squid was not photographed until 2004. Another squid, known as the **colossal** squid, lives in deep Antarctic waters. It is even bigger and harder to find than the giant squid.

Scientists are trying to understand how deep-sea creatures become so enormous in an environment that seems to have a small food supply.

There are many other questions to answer, and many other deep-sea species to discover. Most of the deep ocean is unexplored. It remains a realm of mystery.

LEVEL 20, UNIT 3
INFORMATIONAL TEXT

The Mystery of **THE NAZCA LINES**

Near the coast of northern Chile and southern Peru, between two deep valleys, is the **Nazca Plain**. It is long and narrow, about a mile wide (about 1.6 kilometers) and extending roughly 37 miles (about 60 kilometers). There is almost no wind here and the rains come rarely, maybe once every several years.

This plain is the site of something that has been puzzling the modern world since the 1930s. There are outlines of about 70 animal and plant shapes made by shifting rocks and soil. Many of them are enormous. The largest, a pelican, is about 935 feet long (285 meters). There are also various mathematical shapes, including straight lines, triangles, spirals, circles, and those uneven four-sided shapes known as trapezoids. Some of them are quite sizable, too. The longest straight line extends 9 miles (14.5 kilometers).

Scientists believe they know the who, when, and how of these shapes and lines. It's the why that is the mystery.

Who? The Nazca people **flourished** in this area 1500-2000 years ago. They were an advanced farming culture that built waterways and created beautiful bowls and jars. And they are responsible for these shapes and lines known as the Nazca Lines.

When? Scientists believe that the earliest work was done between 500-200 BCE. The **majority** of the work came later, between 200 BCE and 500 CE.



How? In the earlier works, gravel was often removed and piled inward so that the outlines of the figures were raised above the ground. But mainly, the reddish gravel was carefully removed to uncover the lighter sand underneath. In most regions, this work would barely last a year, brushed away by wind or washed away by rain. But on these dry and windless plains, the Nazca Lines have remained.

Why? Here is a sampling of the many **theories** for the existence of the Nazca Lines.

- Since these lines and drawings can best be seen from the height of a low-flying plane, one **outlandish**, unlikely idea was that they were alien landing strips. (The loose gravel on these plains would have made for very poor spaceship landings.)
- A less unlikely theory suggests that the giant drawings were **offerings** to the gods above, in hopes of receiving good weather for crops.
- It was suggested in the 1940s that the lines might be a guide to the stars, but later studies have challenged this idea.
- One theory links the drawings to religious practices. The straight lines are thought to be roads people traveled to get to important religious sites.
- The mathematical figures seem to match up with actual underground water sources. A theory proposed that they were maps to guide people to water.

No single theory covers all the lines and shapes. It is most likely that the Nazca people had many reasons for creating these fascinating pictures that they left for us to puzzle over.

LEVEL 20, UNIT 4
FOLKTALE

The Maiden Wiser Than the Tsar



A Russian Tale Retold

There was once a powerful **tsar** who ruled with great intelligence. His kingdom almost ran itself, so he was often bored. One day a poor **peasant** came before him seeking employment. The tsar was amazed by this man's gift for speech.

"How is it that you, an uneducated man, are so **eloquent**?" he asked.

"My daughter taught me how to speak so well," the peasant replied.

"She was born wise."

The tsar was immediately interested and decided to test this lady's wisdom. He gave the peasant thirty eggs and commanded, "Tell your daughter to hatch these eggs. If she does, I'll reward you. If she does not, I'll throw you into prison." The eggs, it should be noted, were **hard-boiled**.

The next day, the tsar was making his daily journey down the royal road when he saw the peasant in a nearby field, planting cooked beans.

"What are you doing?" the **astonished** tsar asked.

The peasant smiled **timidly**. "My daughter says that she will hatch cooked eggs, if I can grow plants from cooked beans."

Her cleverness made the tsar more interested, so the next morning, he sent a messenger to the peasant. The messenger handed over a bundle of **flax** and announced, "Your daughter must spin this into cloth and make sails for the tsar's ships by tomorrow or you'll be thrown into prison."

The peasant returned to the tsar that afternoon. Nervously holding up a small block of wood, he said, "My daughter says that if, by tonight, you turn this block of wood into a **loom** to weave the flax, she'll use it to weave the sails by tomorrow."



The tsar could contain his curiosity no longer and had the daughter brought to the palace. Almost as impressed by her beauty as he was by her intelligence, he asked her to marry him and become **tsarina**. The lady had enjoyed the contest as much as the tsar and **consented**—if he would sign an agreement. It stated that if he became displeased with her and sent her away, she would be allowed to take with her the thing she liked best.

Soon after their marriage, the tsar and tsarina got into a silly argument. The tsar demanded that she leave. The lady agreed, but not before making herbal tea for her husband's bedtime drink.

The next morning, when the tsar awoke, he was confused to find himself in the peasant's home. His wife explained that she'd put a sleeping **potion** in his tea. Then she pulled out the agreement he'd signed. "You agreed that when I left, I could take with me the thing I like best," she said, smiling.

The astonished tsar roared with laughter. He and his tsarina returned to the palace and had a long, happy life together.



LEVEL 20, UNIT 5
DRAMA

财富 烦恼
WEALTH AND WORRIES:
A PLAY BASED ON A CHINESE FOLKTALE

Cast of Characters

MR. CHEN, a rich **merchant**

MRS. CHEN, his wife

MR. LI, a poor **laborer**

Scene 1

[Late in the evening in the courtyard of the Chen family's house.]

MR. CHEN is sitting at a table while MRS. CHEN stands nearby.]

MR. CHEN. *[Counting a large pile of gold coins]* Forty-one, forty-two, forty-three, forty-four, forty-five... *[Continues counting as MRS. CHEN comes closer]*

MRS. CHEN. *[Placing her hand on MR. CHEN's shoulder]* My dear husband, you're working much too hard. I am worried about your **well-being**.

MR. CHEN. *[Looking up]* Yes, my hours are long, but I must work hard if we are to remain rich.

[Melody of flute is heard from neighborhood.]

MRS. CHEN. *[Listening]* That's a lovely tune. Neighbor Li must be playing his flute. He knows how to have fun after a day's work.

MR. CHEN. Li works all day digging ditches and chopping wood. He earns pennies. What a terrible way to live!

MRS. CHEN. But isn't that music sweet? Li and his family seem happy, even though they are not rich. We have piles of gold, but are we happy?



MR. CHEN. I'd be happier if I could count these coins in peace without that music distracting me. I have an idea: I'll give Li enough money to make him a rich man. He'll soon be too busy to bother with that flute. First thing tomorrow, I'll have a servant bring Li to me.

[Curtain]

Scene 2

[Early morning in MR. CHEN's courtyard. MR. LI stands respectfully before MR. CHEN, who is seated at a table that holds a small sack.]

MR. CHEN. Neighbor Li, you work so hard yet have no fortune to show for it. I've been thinking of your future. I am giving you the gold pieces in this sack. There is no need to repay me, but you must use it wisely.

[Hands the sack of coins to MR. LI, who looks startled]

MR. LI. My family and I have never had such riches. I am grateful.

[Bows and leaves the courtyard]

[Curtain]

Scene 3

[The same courtyard, three days later]

MR. CHEN. *[Speaking to his wife]* Well, three days have passed since I gave Neighbor Li a sack of gold. My plan worked. He has finally stopped playing that silly flute. Now I can count my gold in peace.

[MR. LI enters holding the sack of gold. He looks exhausted. He bows to MR. CHEN and MRS. CHEN.]

MR. LI. Please forgive me for interrupting you, but I must return this gold. *[He places the sack of gold before MR. CHEN.]* These past few days, I have spent every hour worrying about what to do with such a fortune. I worried about spending it. I worried about making more of it. I worried about someone stealing it. I worried about my children fighting over it. I thank you for the gift, but I must return it.

[MR. LI places a new flute next to the sack of gold.]



MR. CHEN. [*Looking surprised*] What is this?

MR. LI. This flute is my gift to you. Please accept it. When you gave me the gold, I felt the heavy **burden** that you must bear every day. I am most happy when I am making music with my family. I hope that you, too, may find such peace and joy.

[*MR. LI bows and exits. MR. CHEN and MRS. CHEN look confused. Then, MR. CHEN picks up the new flute and begins to play. MRS. CHEN smiles.*]

[*Curtain*]

LEVEL 20, UNIT 6
POETRY

from Songs *for the* People

Journalist, public speaker, activist, and poet Francis Harper was born during slavery in the United States to free parents in 1825. She dedicated her life to African-American rights, women's rights, and the anti-slavery movement. Today, she is known as the "First African-American Protest Poet." In this excerpt from her poem, Harper suggests a way to bring the world together in kindness and peace.

Let me make the songs for the people,
Songs for the old and young;
Songs to stir like a battle-cry
Wherever they are sung.

Our world, so worn and weary,
Needs music, pure and strong,
To hush the jangle and discords
Of sorrow, pain, and wrong.

Music to soothe all its sorrow,
Till war and crime shall cease;
And the hearts of men grown tender
Girdle the world with peace.

- Frances Ellen Watkins Harper

LEVEL 20, UNIT 7
INFORMATIONAL TEXT

The Travels of MARCO POLO

"Here are seen huge **serpents**, ten paces in length.... The jaws are wide enough to swallow a man."

This description of crocodiles comes from one of the most influential books ever written. Its author, Marco Polo, lived from 1254 to 1324.

Marco Polo was born in Venice, a city in present-day Italy that was a leading center of trade. His father and uncle were wealthy merchants and the first **Europeans** to visit the court of **Kublai** Khan, the Mongol ruler of China. When the Polo brothers set off from Venice to visit the Great Khan again, 17-year-old Marco went with them.

The Polos took an overland route. They braved many dangers, from murderous bandits to flooded rivers. They journeyed over mountains and through deserts, finally arriving at Kublai Khan's **court** after a journey of three and a half years.

Kublai Khan took a liking to Marco Polo. The young man spoke several languages and was a sharp observer. Kublai Khan made Marco Polo a trusted **ambassador** and sent him to the provinces of China and beyond, to report back about the lands and customs. The Polos became members of Kublai Khan's court.

When the three merchants returned home to Venice, they had been gone for 24 years.

Venice and its rival city of Genoa fought a war. Marco Polo was captured in battle. He entertained a fellow prisoner with stories of his travels to the Far East. His cellmate, who happened to be a writer, wrote down the stories. After his release from prison, Marco Polo had copies made.



The book, *The Travels of Marco Polo*, described things that were new to Europeans. Marco Polo reported that throughout China, paper money was used instead of metal coins. He said that homes were heated with black stones (coal) instead of wood. He described messengers riding in relays over long distances—a **postal** system.

Polo told about busy cities larger than any in Europe. He described silks, jewels, furs, and other finely made objects. He described elephants on parade and grand **banquets**. He told about the peoples of the empire and how the Great Khan ruled over all.

Did Marco Polo really serve in the court of Kublai Khan? Had he really seen what he described? From his own day to the present, many people questioned the truth of his accounts. It was said that on his deathbed, Marco Polo was urged to **confess** that he had told lies in his book. He claimed, "I have only told the half of what I saw."

Versions of Marco Polo's book were printed in many languages. In the late 1400s, the book inspired another explorer, Christopher Columbus, to set sail on an ocean route to the lands of wealth that Marco Polo so **vividly** described.

LEVEL 20, UNIT 8
INFORMATIONAL TEXT

Dots and Dashes

Dit-dit-dit, dah-dah-dah, dit-dit-dit. Three short, three long, and three short tones are the internationally recognized call for help: S-O-S. Once used by ships and aircraft in trouble, this message is based on Morse code. In Morse code, sequences of dots and dashes stand for numbers, letters, and punctuation. The dots **correspond** to quick tones (dit), and the dashes correspond to longer tones (dah). The sounds were tapped out by **operators** of a **device** called a telegraph.

Early Discoveries

- A scientist in Denmark, Hans Christian Oersted, discovered the connection between electricity and **magnetism** in 1820. A wire carrying an electric current could make a magnetized needle move.
- Oersted's discovery led other inventors to develop devices that used electric **pulses** to make needles point to letters of the alphabet. These were the first telegraphs.
- The word telegraph is based on the Greek word parts tele, meaning "distant," and graph, meaning "writing." In other words, writing that could travel across distances.

A **patent** for the telegraph was given to the American inventor Samuel F. B. Morse in 1840. As with all new **technologies**, Morse's telegraph was based on earlier discoveries. Morse spent years improving the technology, with help from a scientist and an engineer. During that time, he wrote about what the telegraph might achieve, explaining that the telegraph could make "one neighborhood of the whole country." How? By shrinking the time it took to send information back and forth.



Before the invention of the telegraph, communication over long distances depended on mail delivery. Mail traveled across oceans by ship, which could take months. It traveled across land only as quickly as the person, animal, or vehicle that carried it. Morse's telegraph could send pulsing electric signals across wires faster and farther than ever before. It was the first invention that used wires and electricity to send messages, and it would change long-distance communication forever.

How did the telegraph work? A person operating the device sent a message in Morse code using pulsing electric signals. At the receiving end, another skilled operator **decoded** the signals into words. Once the operator translated the message into English, it could be shared.

Telegraph technology required wires strung across land and cables set underwater. After those systems were set up, high-speed, long-distance communication became possible for the first time. Businesses, news reporters, railroads, and the military used telegraphs to send and receive information.

Individuals also sent written messages called telegrams. Telegrams were messages sent by telegraph and then hand-delivered in written form. These communications were limited to special events because they were expensive.

Telegraphs were in use for about one hundred years. Eventually, they were replaced by faster technologies, including telephones. Telephones also relied on wires and electricity, however, they transmitted voices. This difference meant that Morse code was no longer needed: individuals used telephones to talk and listen over long distances.

LEVEL 20, UNIT 9
INFORMATIONAL TEXT

Propaganda or TRUTH?

On February 15, 1898, the American battleship *Maine* blew up in the harbor of Havana, Cuba, killing 266 sailors. But why? Had Spanish forces set off an explosive device? Spain denied it, claiming that the explosion was caused by an accident on board. American newspaper publishers rushed to blame Spain, even without all of the facts. Newspapers urged revenge and called on Americans to “Remember the *Maine*.”

Did Spanish forces actually sink the *Maine*, killing American sailors? Even today, the answer is unknown.

Yellow Journalism: A Brief History

During the late 1800s and early 1900s, strongly expressed opinions without much care for proof were common in American newspapers. This kind of **biased** reporting came to be known as “yellow journalism.” (The name came from “The Yellow Kid,” a popular comic strip in the newspapers.) This type of journalism was designed to offer exciting stories and sell more papers. It was also designed to **influence** people and events.

In the case of the *Maine*, yellow journalism worked dangerously well. In response to what they read in newspapers about how the ship sank, many Americans angrily demanded war against Spain. **Historians** say that yellow journalism played a major role in causing the Spanish-American War.

The Problem: Propaganda

There’s a term for information that gives just one side of an issue and makes people think and act in certain ways: propaganda (PROP-uh-GAN-duh). Often, propaganda supports a **political cause**. Propaganda uses several strategies to influence people and events.



For example, propaganda builds on people's trust in the source of information. In addition, news stories with propaganda include only some facts and leave out others. Propaganda also often **appeals to** human emotions like anger and fear. Lastly, propaganda tends to have slogans that are easily repeated, like "Remember the *Maine*."

Unfortunately, biased reporting in the form of propaganda did not end with yellow journalism of the late 1800s and early 1900s. It still exists today and the problem is worse in some ways. Today, the internet makes instant global communication possible—it's easier than ever to spread the **misleading** ideas and half-truths of propaganda.

One Powerful Solution: Critical Thinking

How can people **resist** the pull of propaganda? They can think critically about what they read and hear. When citizens think critically, they try to understand more than one side of an issue and they look for the truth. They ask themselves questions like,

- Is this a fact or an opinion?
- Are the facts actually true? How can I check?
- Are the opinions based on evidence?
- Why might someone disagree?
- Are there words that quickly make me feel angry or scared?
- Who wrote this? Is the writer likely to be biased?
- Where can I find another point of view?

When people notice propaganda, they can try to think critically and decide for themselves whether to agree or disagree with its message.

LEVEL 20, UNIT 10
NARRATIVE TEXT

Dora, the Dog Wonder

Have you ever noticed that sometimes, as you're getting to know new friends, they do things that surprise you? There might be a kid who sits in the back of the class and acts goofy, but then you hear him singing a solo in the chorus, and you think, "Wow, he's so talented!" Or maybe you meet a kid who hardly ever speaks and seems shy, and then you're **startled** to find out she's earning a black belt in **martial arts**. Something like that happened with my friend Dora last week, and I'm still feeling amazed.

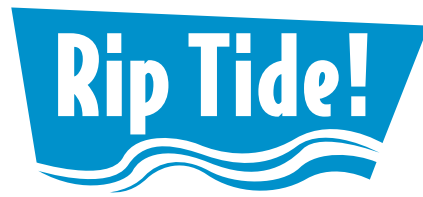
Dora is not a human friend—she's a dog. She came to our family about six months ago, when her owner, my uncle, took a job that required traveling. He couldn't care for her, so we took her in. Dora was the biggest dog I'd ever seen, but she was gentle and enjoyed being near people. My four-year-old sister, May, was always hugging her, and Dora welcomed the **cuddling**. You didn't have to give Dora a command to lie down, because that's what she was doing most of the time anyway. I liked Dora, though she did seem...well...lazy.

Last Saturday my family went to Riverside Park for a picnic, and we brought Dora along. It was the first day without rain in a week, so the river was higher than usual. A family of ducks was on the riverbank, and Mom and May went to look at them.



All of a sudden, I heard Mom screaming. Dad and I jumped up and ran to the water's edge, where we saw Mom thrashing in the wild, **foamy** water. But May was nowhere to be seen. "May fell in!" Mom shouted. Dad quickly leaped into the water, but I just stood there, frozen like a statue, my mind blank with terror. Then, about twenty feet downstream, I saw something **bobbing** in the **swirling** foam. It was Dora's head, and clinging to her neck was May! I hadn't even seen Dora jump into the water. Now I watched her battle the current, swimming powerfully through the moving water until she reached the shore. When she **delivered** May to Mom and Dad, they couldn't stop sobbing with relief.

May was coughing but otherwise fine. I put my arm around Dora and felt how hard she was panting from her lifesaving swim. "What a wonder you are!" I whispered to her again and again and again.

LEVEL 20, UNIT 11
NARRATIVE TEXT

Mira looked up and down the deserted beach, at the choppy water and immense waves, and then at her friend Ashley. "We're here so early, the lifeguards aren't even on duty yet."

"Yes," her friend agreed, "and the beach is empty, so we have it all to ourselves!"

The two girls were visiting Ashley's grandmother, who lived a couple of blocks from the beach, and they had woken up at six o'clock to take an early morning swim.

"I'd feel a lot safer if there were lifeguards watching out for us," Mira fretted.

"But we're not swimming alone," Ashley insisted, "and that's the first rule of swimming safety. Besides, you know we're both good swimmers, and we'll look out for each other."

Ashley tossed her towel on the sand, took off her hoodie, and started trotting toward the surf. Mira frowned and then shrugged, eagerly racing to catch up with Ashley. As the girls dove under an enormous wave that crashed over them, something **bizarre** happened.

When they **surfaced**, both were quite a distance from shore. **Treading water**, the astonished girls looked at each other and then began swimming in, but when Mira looked up to check their progress, she gasped. They were now even farther from shore!

She caught up to Ashley and grabbed her friend's arm to stop her. "What's happening to us?" she sputtered. "It doesn't make sense that we keep getting farther and farther from shore."



Ashley's eyes widened as she recalled a news report she had heard a few weeks ago. "I've heard about something like this. We must be in a rip tide, which is caused by big waves that have created a break in a sandbar offshore. This sets up a fast-moving current that pulls everything in it out to sea, and we're in it!"

Mira felt a sense of **panic** rise up in her throat as she twisted her head around and looked back at the shore. "Oh, no, we're even farther from shore now and we're going to drown!" she moaned desperately.

"We're going to be OK, Mira," Ashley reassured her friend. "We just have to swim **parallel** to the shore, not into shore, because rip currents aren't that big and we'll be able to swim out of it."

Mira nodded, and the girls began swimming. Once they stopped feeling the pull of the current, they turned and headed into shore. Exhausted from swimming, they dragged themselves across the beach and dropped onto their towels. Mira turned and hugged her friend.

"Ashley, you saved us!" she cried gratefully.

Ashley **grimly** shook her head and confessed, "No, I put us in danger by not listening to you in the first place when you wanted to wait for the lifeguards. They would know by looking at the water that something is wrong."

"I've got an idea," said Mira. "When we go home let's get on your grandmother's computer. We can learn more about rip tides!"

LEVEL 20, UNIT 12
NARRATIVE TEXT

A Close Circle of Friends

"Field Day is this Friday," Ms. Kanner told Class 6A during the morning's announcements, "and our class needs four speedy runners for the relay race." She looked around the room. "How about you, Felipe?"

Felipe nodded and **gestured** to his two buddies, Ruben and Jack. Ms. Kanner listed the three names and said, "We need one more."

Nelson felt the **urge** to volunteer because he loved races, but he hesitated. He had been in this school for only two weeks, and it felt as though nobody had even noticed him. Maybe it was because he was small and easy to overlook. Plus, everyone already was part of a circle of friends. Ms. Kanner picked Thomas, whose hand was up, and Nelson tried not to think about his disappointment.

That afternoon Nelson sat on the grass by the track as Felipe, Ruben, Jack, and Thomas practiced running the 400 meters around the oval and passing the tube-shaped **baton**. Thomas had the fastest time, so he was assigned the **anchor leg**. After they left, Nelson jogged around the track a few times, just to loosen up his muscles, before increasing his pace.

On Friday morning, Thomas arrived in school with a **sorrowful expression** and a limp. He explained that his toe had been broken in a skateboarding accident. When Ms. Kanner asked for a replacement for the relay race, Nelson spoke up. "I can run," he offered without hesitating.

Looking at Nelson **skeptically**, Felipe asked, "You sure about that?"

"I can run anchor," asserted Nelson quietly but with confidence.



Felipe ran the first leg of the 1600-meter relay, pacing himself well. He saved a burst of speed for the last 50 meters and then passed the baton to Ruben. Ruben held the lead until the halfway point, when two runners caught up to him. He barely managed to keep up with them. Straining, he reached for Jack's outstretched hand, but there was a **fumble**, and Jack couldn't hold onto the **cylinder**. A dropped baton meant precious seconds lost. By the time Jack had completed his leg and passed the baton to Nelson, Team 6A was in third place.

Nelson eased into the run, keeping his eye on the runner ahead of him. After 100 meters, he pumped harder and passed on the right. At 200 meters, Nelson was on the heels of the first runner, letting her set the pace. "Stay with her, stay with her, stay with her, and GO!" Nelson reached inside for the power he needed. His legs and arms and lungs were in perfect **sync**, and he heard his teammates screaming for him when he crossed the finish line first.

As Nelson leaned over, hands on knees, to catch his breath, Felipe slapped him on the back and laughed, "Man, you're pretty good. What's your name again?"

A few days later, Felipe and his two buddies on the basketball court. "Hey, Nelson, I'm glad you're here!" Felipe called out. "Now we can play two-on-two."

LEVEL 20, UNIT 13
INFORMATIONAL TEXT

The Pollinators

A bee lands on a blossom to sip sweet **nectar** that the hive will use to make honey. The bee's hind legs collect tiny grains of **pollen** to bring back to the hive as food. Other pollen grains stick to its body and drop off at the next flower the bee visits.

Pollen must be **transferred** between male and female plant parts in order for the plant to make seeds and **reproduce**. Wind is one method that plants use to transfer pollen. But 75 percent of the world's flowering plants require animals to do the job. The most common pollinators are bees.

Sweet flower scents and nectars **attract** other insect pollinators, such as flies, beetles, butterflies, and moths. Unlike bees, these insects don't collect pollen on purpose. Like bees, as they travel from flower to flower, their bodies pick up and drop off the sticky grains.

Birds are pollinators, too. A hummingbird hovers above a flower and uses its long bill to sip nectar. When it flies off to the next flower, pollen is clinging to its feathers. Other pollinators are mammals such as bats and small rodents. It's estimated that 100 thousand different animal species pollinate plants throughout the world.

Plants depend on pollinators. People who grow plants, including fruit crops, depend on them, too. Worldwide, some pollinators have been **declining**. There are many causes: disease, **pesticide** use, pollution, loss of habitat, loss of food plants, and more. In an apple-growing region of China, for example, wild bees began vanishing in the 1990s. To keep producing apples, every year thousands of villagers spent days hand-pollinating the apple blossoms with feathered sticks. Eventually, most growers switched to crops that do not need pollination. It was the end of the apple industry in that region.



Everywhere in the world, people can take action to help pollinators. Here are a few ideas.

- DO learn about flowering plants that are **native** to your region and produce a lot of nectar and pollen. Plant a garden with a variety of those plants, so that they flower at different times of the growing season. Even if you live in a city apartment, you can plant window-box flowers to attract pollinators and keep them fed.
- DON'T clear every **brush pile** or cover every bare patch on the ground. They provide materials that bees, birds, and other animals use to build nests.
- DO provide shallow water sources for butterflies and other pollinators, especially when it is hot and dry.
- DON'T try to kill off all the insect pests that eat plants in your garden. Try not to use chemical pesticides to reduce their numbers. Instead, use natural methods, such as including plants to attract predators that eat only harmful insects, not pollinators. Removing pests with gloved hands is another option.

We all need pollinators, and pollinators need us.

LEVEL 20, UNIT 14
INFORMATIONAL TEXT

Most of us don't even notice soil. We call it dirt and make sure to wipe our shoes before bringing that dirty stuff into the house. But as farmers know, soil is precious and essential to life. Without soil, land plants would not grow; without plants, every plant-eater and every living thing that eats plant-eaters would die. Soil is that important! But what is soil?

Little Bits

Soil contains mineral **particles** from rocks that have broken down. If you were to look at a handful of soil from different places, you would see that not all soil is the same. There are differences in color, feel, and the ability to hold onto water. Particle sizes differ, ranging from sand to silt to clay. A good garden soil, called loam, is a mix of sandy, silty, and clay particles. But soil isn't just made of minerals. It also contains vegetable **matter**, such as pieces of leaves and twigs. Animals' bodies provide other **organic** matter. And feasting on all that matter are all sorts of **organisms**. The nonliving, the once-living, and the living are all **components** of soil.

Decomposers

Soil is home to animals that you can see easily, like earthworms and insects, munching on organic matter. If you look closely at a fallen leaf, you may see threadlike **fungi** spreading on it. You won't be able to see the microscopic **bacteria** in soil, though they're amazingly **abundant**. Countless numbers of fungi and bacteria live in soil. A single **gram** of soil can contain thousands of different kinds of living organisms.



Living organisms in soil are called decomposers because their eating breaks down, or decomposes, organic matter. In other words, they make things rot. Over time, the decomposers' actions create humus (HYOO-mus), the rich, organic material that makes the best soil for growing crops. Decomposers in soil provide another essential function: They are nature's **sanitation department**. Without them, the earth would be one gigantic garbage dump, piled high with once-living things that never decay.

Maintaining Healthy Soil

Soil is essential for life on earth. As a result, healthy soil is important for all. But how do we know what makes soil healthy, and how do we keep it that way? Healthy soil does not blow or wash away easily, stores air and water well, and provides **nutrients** for plants. There are a few simple strategies for maintaining healthy soil:

- Leave soil alone—Let decomposers do their work without disruption.
- Grow a variety of plants—Each kind of plant helps soil in a unique way.
- Let roots stay where they are—A plant's roots make nutrients in soil.
- Cover soil up—Soil loves to stay cool and moist.

LEVEL 20, UNIT 15
INFORMATIONAL TEXT

Living with "Good Germs"

We wash our bodies to keep our skin clean. We wash our hands and our food to keep disease-causing germs from getting into our bodies. These are healthy practices, but they don't begin to keep us germ-free. Our bodies are the home planet for an enormous **variety** of tiny creatures, some of them on us and some of them in us. And that's a good thing.

For starters, tiny mites make their homes on our faces and in our eyelashes. Although they are actually relatives of spiders, these creatures are invisible to the naked eye. Each one is less than 0.016 of an inch (0.4 millimeters) long. They can slip easily into facial pores, those extremely small openings on the nose, forehead, cheeks, and chin.

These face mites particularly like to **nestle** inside our **follicles**, the pores from which hair grows. The tiny creatures feed on the oil and dead skin cells in our follicles and also lay eggs there. But don't worry about rising mite populations! Our **immune systems**, the collection of processes in our bodies that protect us from disease, work constantly to keep down these mites' numbers. As a result, we don't even notice them.

Our skin is also home to much, much smaller forms of life, each consisting of just a single cell: bacteria. We generally have over 200 kinds of **bacteria** living on our skin. Instead of harming us, these beneficial bacteria help with the healing of wounds. But that's nothing compared with the bacteria inside our bodies.

A healthy human body contains about 30 to 50 trillion cells—and about that many bacteria, too! Our immune systems target other **microorganisms** that cause us harm but not these bacteria. Like the ones on our skin, these bacteria are basically harmless or even helpful.



Let's start with our mouths. Some 25 kinds of bacteria live around our teeth and in small openings in our gums. Brushing and flossing each day keeps their numbers down and prevents them from forming sticky **deposits** called plaque. But there will always be a large colony of bacteria in our mouths, and that's good. They compete for the same food as the bad bacteria that can get in our mouths, helping to prevent illness.

There are very few bacteria in our stomachs because the stomach is such a high-acid environment. But our guts generally contain at least 500 kinds of bacteria, numbering in the billions. These bacteria help us digest our food (we couldn't do it without them!), produce vitamins, **absorb** nutrients, and fight off harmful bacteria that have invaded our **digestive systems**.

So don't worry about all the extra company our bodies contain. It's perfectly healthy!

LEVEL 20, UNIT 16
INFORMATIONAL TEXT

IT CAME FROM SPACE

Whenever the planet's creatures looked up, they saw a sky filled with familiar things. There was the sun, which appeared and disappeared each day. There was the moon, which grew and shrank, and could bring light to the night. There were clouds, which cast shadows and sometimes brought lightning and rain.

High above the gases surrounding the planet, way beyond the moon, millions of other unseen objects were **orbiting** the sun. These space rocks weren't as large as planets. Many were as small as pebbles, but others were several miles across, even larger. At times, some of these rocky objects came close to the creatures' planet. The planet pulled the objects toward its surface. This pull of **gravity** caused the space rocks to leave their orbit and head toward the planet. Because they were small, they burned up in the **atmosphere** before ever reaching the planet's surface.

But then something happened never before seen by the creatures. Their planet's gravity drew in an object that was unusually large, at least 6 miles wide (10 kilometers). This gigantic object did not **disintegrate** in the air above. It crashed through the atmosphere, burning like a tremendous torch. As powerfully as a fist punching through water, it smashed right into the planet's **crust**. The explosion was deafening.

Anything living near the crash site was wiped out instantly. Creatures far away saw the flaming sky, heard the crash, and felt the heat. Many ran or hid. They had no understanding of what happened and no idea that the sky would soon change and no longer hold its familiar sights.



The **impact** carved out a deep **crater** about 112 miles (180 kilometers) across. Billions of tons of rocks and dust shot into the atmosphere. Fires spread across the land. Smoke and dust blocked sunlight, and endless night came to the planet for months. Without sunlight, plants died. The animals that once ate the plants died, and their **predators** starved, too. Without sunlight, temperatures dropped. The cold killed off animals that needed warmth to live. Smaller animals with fur were better able to survive the cold. They found food in the dead plants and in the seeds that plants left behind. Eventually, the skies would clear, **revealing** a very different planet.

The crash of a giant object from space—an **asteroid**—occurred on Earth 66 million years ago. The event and its effects form the leading scientific theory to explain the **extinction** of dinosaurs and the **rise** of mammals.

LEVEL 20, UNIT 17
MYTH

Phaethon: A GREEK MYTH

Phaethon was just a boy when he **ventured** into the land of the sunrise. The Sun's home could not be visited by **mortals**; however, Phaethon was only half-mortal because his mother was mortal but his father was a god, the Sun himself. Phaethon had never met his father before.

Phaethon entered the Sun's brilliant palace, shielding his eyes from the blazing light. As he approached the Sun's sparkling throne, the Sun asked "What brings you here?"

"My mother tells me that you are my father, but my schoolmates tease me for saying that I am your son. I am here to find the truth," said Phaethon.

The Sun smiled kindly and replied, "Yes, Phaethon, I am indeed your father and now that you are here, I can show my love for you. Ask anything of me, and on my **oath**, you shall have it."

Phaethon recalled how often he had watched his father in his fiery **chariot** flying in a high arc across the sky. It was an amazing sight that made him burst with pride; how thrilling it would be to make that sky journey himself! "I wish to drive your chariot," Phaethon said.

The Sun immediately **regretted** his promise. "I cannot take back my oath," he told Phaethon, "but you do not realize what you are asking. No god but me is strong enough to drive the chariot across the sky. The horses are **mightier** than any you have known, and the dangers are too great. Please, ask for something else."

But Phaethon's mind was made up and he pleaded, "Just for one day." The Sun did not have time for more arguments, for the day's chariot ride was about to begin.



Phaethon stepped into the golden car and took the **reins** in his hands. As the horses galloped out from the gates, their feet flew over the clouds at breathtaking speed. Phaethon held on as the chariot climbed swiftly, but his grip on the reins felt as light as air to the horses. Instead of following their steady path, they swerved to the right and to the left. Out of control, they soared up to the starry heavens, nearly touching the **hideous** fangs and claws of the Scorpion. Phaethon dropped the reins and huddled in terror.

Then the chariot made a steep plunge all the way down to the earth, setting it **ablaze**. The rivers turned to steam, the forests burned to **ash**, and green lands turned into vast deserts. Phaethon, choking on burning smoke, wished for an end to this terrifying ride.

Zeus, the king of the gods, heard Earth crying out for help. He looked down from Mount Olympus and saw the Sun's chariot **careening** wildly. Zeus hurled a thunderbolt at the reckless chariot, causing it to shatter. The horses galloped into the sea, ending the journey.

LEVEL 20, UNIT 18
INFORMATIONAL TEXT

THE MOON:

True or False

How much do you know about Earth's moon? Read these *True* or *False* statements to learn more!

Earth is the only planet in our solar system that has just one moon.

True or False?

The answer is *True*. Neither Mercury nor Venus has a moon. Mars has two moons. All of the giant outer planets—Jupiter, Saturn, Uranus, and Neptune—have multiple moons. Saturn and Jupiter, for example, have at least 53 moons each, with more being discovered.

The moon's craters were mostly formed by volcanoes. True or False?

The answer is *False*. Scientists have found some evidence of volcanic **eruptions** on the moon, but those are not what caused **lunar** craters. Most of the large, bowl-like holes on the moon's surface are impact craters. They formed when **asteroids**, comets, and pieces of those space **bodies** were pulled by the force of gravity and crashed into the moon. The enormous Orientale crater, which can be seen on the lower left edge of a full moon, is more than 600 miles (900 kilometers) wide. Scientists say that it was caused by an asteroid crash about 3.8 billion years ago.

The earth's shadow causes the moon to change shape in phases.

True or False?

The answer is *False*. Watching the moon from Earth, it looks as though the moon changes shape. These "phases" of the moon are not caused by Earth's shadow. Instead, these changes occur because the position of the earth, moon, and sun change each day. Here's more information about why.

When the moon glows at night, the light we see is not from the moon at all. The light is actually sunlight reflected off the moon. Every 29 1/2 days or so, the moon completes a trip around Earth. As the moon's



position changes, different parts of the sunlit moon are **reflected** to Earth. It looks as though the moon is growing from a sliver to a full circle and then shrinking once again. The changing views of the sunlit surface are called the phases of the moon.

- On Day 1, no sunlight is reflected from the moon to Earth; that's the "new moon" phase.
- Next, the moon appears to grow fatter (the growing moon is called a waxing crescent).
- On Day 8, the half-lit moon is in its "first quarter" phase.
- On Day 15, the "full moon" appears.
- On Day 23, the half-lit moon is in its "last quarter" phase.
- We then see a shrinking (waning) crescent.
- On the 29th day or so, we're back to "new moon."

The moon has a "dark side." True or False?

The answer is both *True* and *False*. It takes the moon one month to orbit Earth. As it revolves around Earth, the moon is also spinning, or rotating, on its axis. This rotation is very slow. A lunar day—the amount of time it takes the moon to spin once around its axis—actually takes one month. Because the moon's orbit around Earth and its rotation are the same amount of time, the same side of the moon always faces earth.

Each side of the moon is called a hemisphere. From our **viewpoint** on Earth, we always see the same hemisphere of the moon lit by the sun. We never see the other hemisphere when it is lit by the sun. In other words, viewed from Earth, the moon does have a "dark side" because it looks as though one side of the moon is always dark. However, the sun does shine on both moon hemispheres. Therefore, if we could watch the moon from the sun, we would see both sides of the moon illuminated: neither side would be dark all of the time. Viewers from Earth just never get a glimpse of the other side. That's why the answer is *True* (the moon looks like it has a "dark side" when viewed from Earth) and *False* (both sides of the moon reflect light when viewed from the sun).

LEVEL 21, UNIT 1
NARRATIVE TEXT

TRAVELS WITH RAY

Hey there, Earthling, it's me Ray, just an ordinary ray of sunlight traveling at the speed of light down to your planet, Earth. I may be moving fast, but I have a lot of universe to cover, so we have some time to get to know each other. (Eight minutes and 20 seconds, to be precise.)

I was born on a giant ball of hot gas that you Earthlings call the sun. And believe me, everything your science teacher has told you about the sun is true. It's hot, it's bright, and you definitely wouldn't want to live there.

All my life, I've known I was **destined** for something big, something life-changing, more than just day after day of blazing heat and light. I've always had a lot of energy, so when I heard that sunlight supplies a whole system of **transferring** energy on Earth, I jumped at the chance to have that kind of power. Seriously, I jumped, and here I am—zooming through the **atmosphere**, about to reach Earth's surface.

Crikey, is that the fresh scent of Earth air? Do I feel those dainty drops of moisture you Earthlings call rain? Yes, indeed, I've finally arrived and now it's my time to shine. Those plants over there are such a spectacular shade of green, but—watch out—I'm going to crash!

Oh boy, what a **colossal** mess I've made of my first assignment on Earth. Wait a minute, something strange is happening to me....

Pssst, I'm over here, inside this green plant. After I crashed into one of its leaves, I got tangled up with some **molecules** in here, which meant using some of my energy, and now I just don't feel like myself anymore. Oh well, this might be my only chance to make some friends here on Earth, so I may as well look around.



Awww, look at that cute little mouse. Poor thing, it looks so hungry. Hey there, mouse, stop eating my leaf! Oh my, now it's consuming me, and something very strange is happening again...

Hello again! It's me, Ray, inside the mouse. After it ate that leaf, I was thrown together with more molecules, and now I feel completely different with even less energy than before. This must be what it's like to be part of a food web—Earth's system of transferring energy from one **organism** to another.

Don't look now, but I think I see a snake, and—if I'm not mistaken—snakes eat mice. I'm giving the mouse energy so it can escape, but I don't think it's enough. Oh dear, the snake is eating my mouse... and me. Energy is transferred again! What on Earth will happen next?

LEVEL 21, UNIT 5
NARRATIVE TEXT

This is an excerpt from a longer story, “Sol Painting, Inc.,” by Meg Medina. In this part of the story, Merci Suarez wonders about starting seventh grade at the school her older brother attends. Merci wishes she could spend the summer helping her father with his painting business, but her mother has other plans.

I wonder what it’s going to be like for me this year. Roli and I will both be at Seaward Pines, although I’ll be in the lower school with all the other seventh-grade “**amoebas**” (his word). Roli is five years older than I am—a senior. The last time we were in the same school, I was in kindergarten and he was one of the bossy safety patrols with a plastic badge. After that, he became a Sunshine Scholar at fancy Seaward Pines School, where everybody thinks he’s a genius.

Mami says I’m going to love Seaward Pines, but I don’t know. I’m not much for fancy, and everything about that place is shiny and stiff. Even the red blazers I’ll have to wear look hot and silly, if you ask me. Plus, no one from our neighborhood goes there, except Roli, so I’ll have to make new friends. Stuff like that doesn’t bother Roli. In fact, he’s never brought home a friend in all the years he’s gone there. I asked him about it once, but he told me to close my **oral cavity**.

I think what Mami really means is that she’s going to love it. Last year was tough on her. My highest grade was a C.... Well, it was frustrating for me, too. To think, all my years of perfect attendance and neat penmanship did absolutely nothing to butter up my teachers at report card time. It’s what we call a poor **return on investment** in the business world.



Mami finally said, “**iHasta aqui!**” and called Papi to “discuss my future,” so I knew I was dead meat. I fought it as best I could, but they decided that I needed “a more structured learning environment,” **aka** Seaward Pines.

“Why does it matter if I get an A in science or English?” I cried to Mami. “I’m going to take over Papi’s business anyway!”

She gaped at me like a fish out of water. “Business? Is that what you’re calling a dented van and the few guys who show up when they feel like it? A business?”

Mami: She has no vision. No wonder she and Papi don’t get along.

Anyway, with Roli’s help, I managed to **broker** a deal. I agreed not to run away. I’d go to Seaward Pines but only if I could **apprentice** with Papi and get paid. So far, they’re living up to the agreement. I’m twelve, so for now, I mostly do the trim, and I’m not allowed to go on all the jobs on account of child labor laws and all that [nonsense]. I’ve been on two sites so far: Ramon’s Auto Parts (not bad since it was air-conditioned) and the **marina**, which left me smelling like bait for days.

LEVEL 21, UNIT 6
NARRATIVE TEXT

Secret Samantha

Excerpt from a Story by Tim Federle

This is an excerpt from a longer story, "Secret Samantha," by Tim Federle. In this part of the story, Samantha needs to find a gift for Blade, a new girl in school whose name she drew from a hat to be her Secret Sharer. It's supposed to be a fun holiday activity, but Sam and her mom can't agree on what to buy.

The mall is a zoo, if the zoo forgot to build cages.

Half the shops are permanently closed, and the food court is a war zone. But after we pick up [a gift card] for Miss Lee, Mom **beelines** straight for a boutique that's within the price range of our Secret [Sharer] rules.

"What about this, Sammy?" she says, holding up a mini makeup kit that's right by the front of this **quirky** pop-up shop. Mom calls me Sammy because she knows I hate Samantha, and she knows I prefer Sam, and so Sammy is kind of "in the middle." We are working on being in the middle with one another.

"Maybe," I say, and Mom goes, "Speak up, Sammy," and I say, "Maybe."

That right there is a good example of me being in the middle, because I wanted to throw the makeup against the wall and shout, "Blade doesn't need makeup to be cool!"

"Well, I'll hold on to it," Mom says, making apology eyes at the cashier. "Unless you find something better."



But I don't find something better, is the problem. I go up and down the aisles, and I find things that I could use—like these **turquoise** dragon earrings that might distract people from my giant ears, or this mini **embroidered** DIVA pillow that would be a good bed for my bunny, Sir Hop-a-Lot—but none of it seems very Blade.

"Samantha?" I hear from the front. I'm allowed out of my mom's sight for about ten seconds for every year I've been alive.

As I'm shuffling back to her, I'm making note of the silver high heels she's got on that I know make her feet "scream." She always looks so **dolled up** but so uncomfortable. She's the very opposite of me. I'd rather be plain and relaxed. If I could wear sweatpants to church, I would. If I could buzz off my hair, I would. I'm serious!

"So?" Mom says, tapping the makeup kit against the counter. "Shall we?"

But then she gets a phone call from her boyfriend—I recognize the special ring—and she hands the cashier both of her credit cards, says, "One of those should work," and steps away to take Scott's call.

LEVEL 21, UNIT 9
INFORMATIONAL TEXT

Lonnie Johnson,

Innovating from the Beginning

Lonnie Johnson is always thinking about how the world works and how it can work even better. Ever since he was a boy in Alabama, USA, Lonnie has been an **innovator**. His natural curiosity led him to a career in **engineering** and invention that has been a roller coaster ride of both successes and hard times.

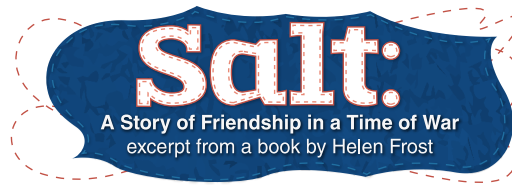
Growing up, Lonnie's family encouraged him to be creative and follow through on his ideas. He designed go-karts, made rocket fuel, and even built a robot from scrap metal. Lonnie named the robot Linex and entered it in a science competition. He won first prize!

Lonnie received a scholarship to study at Tuskegee University where he earned a degree in engineering. After graduating, Lonnie joined the United States Air Force. He then worked at the National **Aeronautics** and Space Administration (NASA) on the space program. He built and tested spacecraft models with a team of engineers. Lonnie was successful and well-respected, but he had always dreamed of being a famous inventor.

Lonnie left NASA to work on his own inventions full time. Unfortunately, he couldn't convince companies to produce any of his ideas. Without a job and without a successful product, Lonnie and his family did not have enough money. They were forced to move out of their home into a much smaller apartment. Life was hard—much harder than when Lonnie worked at NASA or with the Air Force. Anyone else might have given up.



But Lonnie refused to let go of his great ideas. “I don’t know what it is with me, but I’ve never been very good at giving up,” Lonnie has said. So, despite many **setbacks**, he kept improving his designs. It took seven years, but eventually Lonnie found the right toy company for his most famous **innovation**, the Super Soaker™ water toy. Never before had a water toy been able to squirt water so far. Now he spends his days inventing **technology** that will help the planet.

LEVEL 21, UNIT 14
POETRY

Salt: A Story of Friendship in a Time of War, by Helen Frost, tells a story from the different **perspectives** of two friends at the beginning of the War of 1812. Both boys are twelve years old. James is a white settler living his family at Fort Wayne in what is now Indiana, USA. Anikwa is Native American and lives in a nearby village with his family and other members of the **Miami** nation. In this excerpt, Fort Wayne is under attack, and the boys are thinking about how the war has impacted their lives.

ANIKWA

It's raining.

Like everyone in **Kekionga**,
we've invited people we don't know
to stay with us. The man we saw
standing in the hole is here.

He's **Ojibwe**—Father
remembers him

from the Greenville treaty-signing,
and tells us the man's name is "Brings In Light."

It's true—light from the fire bounces off his face and shines on us.

Wedaase is here, too, sitting beside Father, telling everyone about the time
he saw me with James. He asks if I'm a spy for the Americans. (Does he mean it?)

My face turns hot, and Father answers for me: *We trust James and his family.*

*His mother has always been kind to us. She took care of Anikwa's mother
when we were too sick to care for her ourselves. Ever since, we've
called her Sister. James is like a cousin to our children.*

*Wedaase says, Be careful, friend. That kind
of cousin can turn his back on you
when you need him most.*

Father looks at Wedaase, then at me.

Is he remembering the kindness Mrs. Gray showed
my mother long ago, or is he thinking about the other day,
when Mr. Gray looked at the floor and said,

*No more **salt**?*

**JAMES**

Oatmeal for lunch again? Two raisins each for me and Ma, three for Pa. Same as we had for lunch and breakfast yesterday! I'm sick of oatmeal.

Why can't we have bread and cheese for lunch? I ask. Ma looks at Pa. *Don't complain*, he says. (I didn't complain—I only asked.) *We have to make*

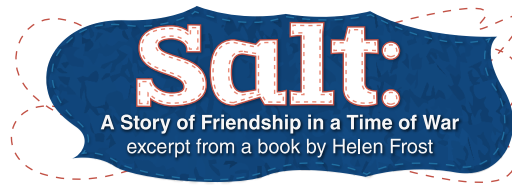
*our **provisions** last.* (For how long?) He's already told me we can't go out hunting or fishing "until this is over with." I look at my feet, and Molly's,

thinking about the moccasins Mink makes, and the socks Ma knits for our Miami friends. Not just trading, more like friends or relatives.

Ma's being quiet. Could I talk to Pa about what Isaac said? *Pa*, I say, *Isaac says the Indians are on the British side. I thought they were on ours.*

He answers: *They're coming here from all around—we don't know them all. It's hard to say who our friends are.* Ma looks out the doorway. Dark clouds

are gathering behind the flag. *As long as we have no evidence to the **contrary**,* she says, *we'll continue to treat the Miami as the friends they've always been.*

LEVEL 21, UNIT 15
POETRY

Salt: A Story of Friendship in a Time of War, by Helen Frost, tells a story from the different **perspectives** of two friends at the beginning of the War of 1812. Twelve-year-old James lives with his family in a military fort in what is now Indiana, USA. Also 12 years old, Anikwa lives in a nearby village with his family and other members of the **Miami** nation. In this excerpt, American soldiers have burned Anikwa's village in revenge for a Native American attack. The boys and their families get together for the first time after the fire.

ANIKWA

It's almost dark
when we walk into **Kekionga**.
Or where it used to be. Now it's ... ashes.
Kwaahkwa told us, but no one could
imagine how terrible it is:
every house
torn apart
and burned. The fish
we had to leave on drying racks
scattered everywhere, and trampled.
Corncobs and fish heads covered with flies—
the army must have eaten what they wanted, and then
destroyed whatever was left over. Didn't they know—
they must have known—it's too late to grow more
corn, and we won't be able to catch many
fish before the river freezes.
Will the animals find
their way
back?
Will deer give us
hides for warmth and shelter, meat
for winter food? Father says, *It's worse than I thought*.
Grandma says, *You helped them go to **Piqua**.*
They should help us now.

**JAMES**

Anikwa looks at me like he's forgotten who I am, his eyes so sad and angry I don't know what to say. I made him a new whistle, but now

it seems like it belongs someplace we can't go back to. I keep it in my pocket. We give them a ball of twine, a new blade for their saw.

Ma says, *This is not on **credit**. It's a gift.* Pa looks surprised to hear that, but he doesn't disagree. A few leaves are turning yellow, falling

from the trees into the river. When I do offer the whistle to Anikwa, he takes it, but he doesn't smile. He looks older. He looks hungry.

They all do. **Mink** spreads out her hands and speaks to us. **Old Raccoon** repeats her words in English: *Please sit with us and eat.* He goes away and

comes back with a roasted rabbit. Mink holds out a small gourd bowl to Pa. He turns red and looks down at the ground before he dips his fingers in. **Salt.**

Thank you, he says softly. I whisper to Ma, *They hardly have any food. Why are they feeding us?* She answers, *You should know by now. This is who they are.*



Glossary

ablaze (adjective) Ablaze means on fire.

absorb (verb) To absorb is to soak up.

abundant (adjective) Abundant means in large amounts.

aeronautics (noun) Aeronautics is a branch of science that studies air travel.

agonizing (adjective) Agonizing means extremely difficult.

aka (acronym) Aka means *also known as*.

algae (noun, plural) Algae are basic water plants without stems or leaves.

alien (adjective) Alien means strange and unfamiliar.

ambassador (noun) An ambassador is a person sent to a country as a representative.

amoeba (noun) An amoeba is a one-celled organism.

anchor leg (noun) The anchor leg is the final section of a relay race.

antennae (noun, plural) Antennae are long, thin body parts used for sensing.

appeals to (verb, phrase) To appeal to is to be pleasing or interesting to someone.

apprentice (verb) To apprentice means to work while learning a skill.

arachnid (noun) An arachnid is an animal with eight legs, like a spider.

Argonaut (proper noun) An Argonaut is one of the heroes who sailed with Jason.

aroma (noun) An aroma is a pleasant smell.

ash (noun) Ash is a powdery substance left after a fire.

assigned (adjective) Assigned means given to.

asteroid (noun) An asteroid is a rocky object circling around the sun.

astonished (adjective) Astonished means greatly surprised.

atmosphere (noun) The atmosphere is the layer of gases surrounding the planet.

attract (verb) To attract is to cause something to come closer.



axis (noun) An axis is an imaginary line that something, like the Earth, spins around.

bacteria (noun, plural) Bacteria are tiny living things made of one cell.

banquet (noun) A banquet is a large, fancy meal.

baton (noun) A baton is a short stick.

beeline (verb) To beeline means to move toward in a straight line.

Beethoven (proper noun) Ludwig van Beethoven is the name of a famous composer.

befriend (verb) To befriend is to become friends with.

biased (adjective) Biased means unfairly preferring one group to another.

bison (noun) A bison is a large, shaggy mammal.

bizarre (adjective) Bizarre means very strange.

bob (verb) To bob is to bounce up and down.

body (noun) A body is an object set apart from other objects, like the moon.

bonus (noun) A bonus is an extra reward.

broker (verb) To broker means to arrange or organize.

brush pile (noun) A brush pile is a stack of broken twigs or branches.

budget (noun) A budget is a plan for how to spend money.

burden (noun) A burden is something that is difficult or worrisome.

burr (noun) A burr is a prickly covering of a nut or seed.

burrow (noun) A burrow is a hole or tunnel where an animal lives.

captivity (noun) Captivity is the state of being confined.

careen (verb) To careen is to sway dangerously from side to side.

carving (noun) A carving is an object cut from a hard material like wood.

chariot (noun) A chariot is a kind of cart pulled by horses.

citizen (noun) A citizen is a resident of a particular place.

clarinet (noun) A clarinet is a long, straight musical instrument.



clog (verb) To clog is to block the flow of something.

colossal (adjective) Colossal means extremely large.

comparatively (adverb) Comparatively means compared to something else.

component (noun) A component is a part.

compromise (verb) To compromise means to reach an agreement by working together.

conduct (verb) To conduct is to transfer, for example, heat or electricity.

confess (verb) To confess means to admit the truth.

consent (verb) To consent is to agree to something.

conservation (noun) Conservation is the act of protecting a natural resource.

contrary (noun) Contrary means the opposite.

correspond (verb) To correspond is to match with something.

court (noun) A court is a group of advisers that works for a queen or king.

crater (noun) A crater is a hole in the ground made by an object falling.

credit (noun) Credit is a loan of supplies until payment is made.

cross (adjective) Cross means angry or bothered.

crust (noun) A crust is the hard surface layer of something.

cuddle (verb) To cuddle is to snuggle.

current (noun) A current is a body of air moving in one direction.

curtain (noun) In plays, curtain means the end of a scene.

cylinder (noun) A cylinder is an object shaped like a tube.

decline (verb) To decline is to decrease in number.

decode (verb) To decode is to change into words.

deed (noun) A deed is a bold act.

deliver (verb) To deliver is to bring and hand over.

deposit (noun) A deposit is a layer or clump of some material.



destined (adjective) Destined means meant to be.

device (noun) A device is a machine that does a special job.

devour (verb) To devour is to eat quickly and hungrily.

digestive system (noun) A digestive system is the set of processes that breaks down food inside the body.

disintegrate (verb) To disintegrate is to break up into small pieces.

district (noun) A district is an area or neighborhood.

doggy-paddle (noun, used as adjective) A doggy-paddle is a swimming movement like a dog.

dolled up (adjective) Dolled up means dressed in fancy clothes.

drought (noun) A drought is a long period without rain.

eloquent (adjective) Eloquent means clear and persuasive in speaking.

embroidered (adjective) Embroidered means decorated with a pattern of thread.

emerge (verb) To emerge means to appear from somewhere.

encounter (verb) To encounter is to meet or find.

engineer (noun) An engineer is a person who designs buildings or machines to solve problems and make life better.

engineering (noun) Engineering involves designing buildings or machines to solve problems or make life better.

epicenter (noun) An epicenter is the point at the very center of something.

erupt (verb) To erupt is to burst out suddenly.

eruption (noun) An eruption is when something explodes out of something else.

European (proper noun) A European is a person from Europe.

evacuate (verb) To evacuate is to remove people from danger.

excerpt (noun) An excerpt is a small part of a longer story.

exotic (adjective) Exotic means unusual or coming from a faraway land.



expression (noun) An expression is the look on someone's face.

extinction (noun) Extinction is when an entire type of plant or animal has died off and become extinct.

flax (noun) Flax is a kind of plant used to make fabric.

flourish (verb) To flourish is to thrive and grow in a healthy way.

foamy (adjective) Foamy means sudsy.

follicle (noun) A follicle is a pore from which hair grows.

force (noun) A force is an energy or power.

foresee (verb) To foresee is to see into the future.

fumble (noun) A fumble is a clumsy move.

fungi (noun, plural of fungus) A fungus is a living thing, like mold or mushrooms, that eats decaying material.

furious (adjective) Furious means powerfully fast or violent.

gesture (verb) To gesture is to make a motion that shows feelings or ideas.

glisten (verb) To glisten is to shine or sparkle.

gram (noun) A gram is a small unit of weight.

gravity (noun) Gravity is a force in nature that brings objects towards each other.

grimly (adverb) Grimly means in a serious way.

habitat (noun) A habitat is the place where a plant or animal usually lives.

hard-boiled (adjective) Hard-boiled means cooked until solid.

hemisphere (noun) A hemisphere is one half of a round object.

hesitate (verb) To hesitate is to stop for a moment when unsure.

hide (noun) A hide is the skin of an animal.

hideous (adjective) Hideous means unpleasant to look at.

historian (noun) A historian is a person who studies events in the past.

hoof (noun) A hoof is a horse's foot.



iHasta aqui! (phrase) iHasta aqui! means *I'm fed up!*

immune system (noun) An immune system is the set of processes that protect a living thing from disease.

impact (noun) An impact is the force of an object crashing into another object.

influence (verb) To influence is to have an effect on someone or something.

innovation (noun) An innovation is a new idea, tool, or way to do something.

innovator (noun) An innovator is a person who has new ideas, inventions, or ways of doing things.

jutting (adjective) Jutting means sticking out.

Kekionga (proper noun) Kekionga is a Miami village near Fort Wayne.

Kublai (proper noun) Kublai is the name of a person.

Kwaahkwa (proper noun) Kwaahkwa is one of Anikwa's friends.

laborer (noun) A laborer is a person who is paid to do hard physical work.

loom (noun) A loom is a tool used for making fabric.

lunar (adjective) Lunar means having to do with the moon.

magnetism (noun) Magnetism is the power attracting two metals or magnets.

majority (noun) The majority is the larger amount or number.

marina (noun) A marina is a place to keep boats on an ocean or lake.

martial arts (noun, plural) Martial arts are self defense skills.

mate (noun) A mate is a partner.

matter (noun) Matter is a certain kind of material.

melody (noun) A melody is a musical tune.

merchant (noun) A merchant is a person who sells things.

Miami (proper noun) Miami is a Native American nation located in what is now Illinois, Michigan, Indiana, and Ohio.



microorganism (noun) A microorganism is a living thing so tiny it must be viewed with a microscope.

mightier (adjective) Mightier means more powerful.

Mink (proper noun) Mink is a woman who adopted Anikwa after his parents died.

misleading (adjective) Misleading means causing someone to believe untrue information.

molecule (noun) A molecule is the smallest part of something that is still similar to it.

monitor (verb) To monitor is to observe and keep track.

mortal (noun) A mortal is a being who does not live forever.

murmur (verb) To murmur is to make a quiet or gentle sound.

musical (noun) A musical is a movie or play that tells a story through songs.

muster (verb) To muster means to work hard to find.

mutineer (noun) A mutineer is a soldier or sailor who disobeys orders.

native (adjective) Native means naturally growing in a particular place.

Nazca Plain (proper noun) Nazca Plain is the name of a region.

nectar (noun) Nectar is a sweet liquid that plants make.

nestle (verb) To nestle is to cuddle.

nutrient (noun) A nutrient is a substance that helps plants and animals live and grow.

oath (noun) An oath is a strong promise.

offering (noun) An offering is a gift.

Ojibwe (proper noun) Ojibwe is a Native American nation.

Old Raccoon (proper noun) Old Raccoon is a man who adopted Anikwa after his parents died.

operator (noun) An operator is a person who works a machine.

oral cavity (noun) An oral cavity is a mouth.

orb (noun) An orb is a rounded shape or a sphere.



orbit (verb) To orbit is to circle around something.

orchestra (noun) An orchestra is a group of musicians who play together.

organic (adjective) Organic means from a living thing.

organism (noun) An organism is any living thing, such as a plant, animal, or fungus.

orphan (noun) An orphan is a child or animal that has no parents.

outage (noun) An outage is a period when a service is unavailable.

outlandish (adjective) Outlandish means very strange.

outwit (verb) To outwit is to win by using more clever ideas.

overhear (verb) To overhear is to listen in on someone's conversation.

pace (verb) To pace is to walk back and forth when anxious.

panic (noun) Panic is a feeling of extreme fear.

parallel (adjective) Parallel means in the same direction when side by side.

paralyzed (adjective) Paralyzed means unable to move.

particle (noun) A particle is a very small piece of something.

patent (noun) A patent is an official document for a new invention.

peasant (noun) A peasant is a farm worker who has little money.

perspective (noun) A perspective is a way of looking at and understanding something.

pesticide (noun) A pesticide is a substance used to kill insects.

Phaethon (proper noun) Phaethon is the name of a character in Greek mythology.

pickpocket (noun) A pickpocket is a person who steals from pockets.

pinch (noun) A pinch is the small amount of something held between a finger and a thumb.

pint (noun) A pint is a unit used for measuring a liquid.

Piqua (proper noun) Piqua is a town. Anikwa's father protected many families when they went to Piqua during the war.



plover (noun) A plover is a small bird that wades in the sea.

political cause (noun) A political cause is a goal or a belief that politicians support.

pollen (noun) Pollen is tiny plant dust, usually yellow, that helps make seeds.

postal (adjective) Postal means related to mail delivery.

potion (noun) A potion is a liquid mixture with special properties.

predator (noun) A predator is an animal that hunts and eats another.

pressure (noun) Pressure is a force pushing on something.

prey (noun) Prey is an animal that is hunted and eaten by another.

propel (verb) To propel is to push forward.

Psammead (proper noun) Psammead is the name of the type of creature the children meet in the story.

pulse (noun) A pulse is a short increase in the amount of something, like electricity.

quest (noun) A quest is a long trip to find something.

quirky (adjective) Quirky means unusual.

ram (noun) A ram is a male adult sheep.

reaction (noun) A reaction is a response to something.

realm (noun) A realm is a large region.

reassure (verb) To reassure means to make someone feel less worried.

rec center (noun) A rec center, also known as recreation center, is a place where people gather for activities.

reduce (verb) To reduce is to make smaller.

reflect (verb) To reflect is to send something, like light, back.

regret (verb) To regret is to feel sorry about something.

reins (noun, plural) Reins are long straps used to guide a horse.

repel (verb) To repel is to keep something from entering or sticking.

reproduce (verb) To reproduce is to make new life.



resist (verb) To resist is to stay strong against the effect of something.

return on investment (noun) A return on investment is money earned by a business beyond an amount that was spent to start it.

reveal (verb) To reveal is to tell or show something.

revolve around (verb) To revolve around is to move around a path shaped like a circle.

rousing (adjective) Rousing means exciting.

rubbish (noun) Rubbish is trash.

salt (noun) Salt is a mineral used to make meat last and add taste. In 1812, salt was traded for food and supplies.

sanitation department (noun) A sanitation department is a group responsible for keeping places clean.

savagery (noun) Savagery is acting in a way that is cruel or damaging.

scheming (adjective) Scheming means crafty or likely to make secret plans.

scornfully (adverb) Scornfully means in a way that shows strong disapproval.

sensation (noun) A sensation is a feeling.

serpent (noun) A serpent is a large snake.

setback (noun) A setback is a problem that gets in the way of reaching a goal.

shudder (verb) To shudder is to shake from fear.

simultaneously (adverb) Simultaneously means happening at the same time.

skeptically (adverb) Skeptically means with doubt.

sorrowful (adjective) Sorrowful means extremely sad.

span (noun) A span is a limited length of time.

species (noun) A species is a kind of plant or animal.

startle (verb) To startle is to surprise someone.

startled (adjective) Startled means surprised.



stringed instrument (noun) A stringed instrument is a musical instrument with strings, like a violin.

structure (noun) A structure is the way parts are arranged.

surface (verb) To surface is to come to the top.

surface (noun) The surface is the outer layer.

surge (verb) To surge is to move forward powerfully.

swiftly (adverb) Swiftly means quickly.

swirl (verb) To swirl is to spin or whirl.

symphony (noun) A symphony is a kind of musical composition.

sync (noun) To be in sync is to be in perfect rhythm.

technique (noun) A technique is a way of doing a task.

technology (noun) Technology is a tool invented to make life easier.

Thailand (proper noun) Thailand is the name of a country.

theory (noun) A theory is an idea based on facts that might explain something.

threatened (adjective) Threatened means in danger.

timidly (adverb) Timidly means nervously or without courage.

torment (verb) To torment is to cause pain.

transfer (verb) To transfer means to move from one place to another.

transportation (noun) Transportation is something that carries people and things from one place to another.

tread water (verb) To tread water is to swim in place.

trombone (noun) A trombone is a musical instrument made of metal.

tsar (noun) A tsar is a ruler similar to a king.

tsarina (noun) A tsarina is a ruler similar to a queen.

Tunguska (proper noun) Tunguska is the name of a river.



turquoise (adjective) Turquoise is a light blue-green color.

twilight (noun) Twilight means the time between day and night.

unidentified (adjective) Unidentified means unknown.

urge (noun) An urge is a strong desire.

vanish (verb) To vanish is to disappear.

variety (noun) A variety is an assortment of many different things.

vegetation (noun) Vegetation is the group of plants in an area.

vehicle (noun) A vehicle is something used for carrying people or goods.

venture (verb) To venture is to go on a journey.

vessel (noun) A vessel is a ship.

viewpoint (noun) A viewpoint is where you are when you look at something.

vividly (adverb) Vividly means colorfully or with great detail.

Wedaase (proper noun) Wedaase is a Native American man from the Ottawa nation.

well-being (noun) Well-being is the state of being happy and healthy.

wildfire (noun) A wildfire is a quick-moving fire in nature that is hard to stop.

willingness (noun) Willingness means a state of being ready.

wilted (adjective) Wilted means limp and faded.

witness (noun) A witness is a person who sees something happen.